

Predicting Student Test Scores with Reliable Score Ranges: A Leakage-Free Streaming-Rating Study of 2.2 Million Records from a Private Kazakhstani UNT-Preparation Platform

Abylay Salimzhanov^{1,*}, Artem Bykov², Aadm Ziro²,
Ruslan Bulgakov³, Aasso Ziro^{1,3}

¹School of Information Technologies,

Kazakh-British Technical University (KBTU), Almaty, Kazakhstan

²International Information Technology University (IITU), Almaty, Kazakhstan

³Al-Farabi Kazakh National University (KazNU), Almaty, Kazakhstan

*Corresponding author: a.salimzhanov@kbtu.kz

ORCID: 0000-0001-6630-585X

Abstract

Reported accuracies in student performance prediction are frequently inflated by feature leakage and non-chronological evaluation, and point forecasts give educators no signal of when a prediction should not be trusted. We address both problems on 2,208,313 practice-test records from 55,482 students (2018–2025) on a Kazakhstani university-entrance-preparation platform. A factorial audit shows that two common feature-engineering errors combined with random splitting inflate the coefficient of determination by 0.349 relative to a leakage-free chronological protocol. We propose **GRACE** (Glicko-informed Rating-Augmented Confidence Estimation), a streaming rating layer that maintains per-student ability estimates and per-content difficulty estimates with a rating deviation that grows with inactivity and shrinks with each observation. The state is computed in a single chronological pass, making target leakage structurally impossible. The rating features feed an XGBoost + LightGBM gradient-boosted ensemble—compared against Ridge regression and Random Forest under controlled conditions, and a single-layer LSTM as a reference data point—that outputs a predicted score together with a 90% score range whose width adapts to measured uncertainty. GRACE attains $R^2 = 0.556$ (mean absolute error 11.72 percentage points) against 0.444 for the leakage-free baseline, with stable out-of-time performance across five annual windows ($R^2 = 0.522 \pm 0.052$ versus 0.288 ± 0.151 for the aggregate baseline). Score ranges closely approach the 90% nominal coverage target (0.886 empirical, 1.4 pp below), with mean width 48.0 pp; coverage is consistent across RD strata (0.880–0.891). In a controlled comparison holding the feature set and training data constant, gradient-boosted trees outperform bagging (Random Forest: $R^2 = 0.517$) and linear models (Ridge: $R^2 = 0.489$) by confirmed margins. The rating layer updates in constant time without retraining, making honest, cold-start-aware prediction practical for operational platforms.

Keywords: educational data mining; student performance prediction; knowledge tracing; Elo rating system; prediction intervals; data leakage; gradient boosting; out-of-time evaluation; conformal prediction; Unified National Testing

1 Introduction

Quantitative prediction of student academic performance is a core task of educational data mining (EDM), enabling personalized instruction, early identification of at-risk learners, and evidence-

based management of educational institutions [1, 2, 3, 4]. Two obstacles limit the practical value of reported accuracy. The first is evaluation validity. Leakage of future outcome information into features, and the use of random rather than chronological splits, together drive a reproducibility crisis in ML-based science [5]; yet chronological evaluation remains rare in the student-performance literature [6, 7]. The second obstacle is uncertainty quantification. A point forecast gives an educator no indication of how much a prediction can be trusted—a limitation that is most acute for students with little history or long inactivity, precisely the situations where reliable guidance matters most.

In Kazakhstan, the Unified National Testing (UNT) serves as the principal gateway to university admission and state educational grants [8, 9], generating high demand for private test-preparation platforms. Machine learning applications to education in the region remain small-scale and institution-specific [43]. No prior work has combined leakage-audited evaluation with reliable score ranges on longitudinal platform data at this scale from Central Asia.

This study operates at a scale—2.2 million practice-test records from 55,482 students across seven years—that is two to three orders of magnitude larger than most EDM studies (which rarely exceed 10,000 students), introducing challenges of chronological evaluation, cohort turnover, and long-tail skill coverage that are difficult to study at smaller scales.

This paper makes four contributions.

1. **GRACE: a hierarchical streaming rating system for continuous educational outcomes** in which the rating deviation serves a dual purpose: it acts as a real-time uncertainty state within the rating layer *and* as the pre-specified grouping variable for Mondrian conformalized quantile regression, so that prediction-interval width adapts automatically to measured uncertainty. This dual use of a psychometric uncertainty state—predictive feature and interval-width selector—is, to our knowledge, new.
2. **Asymmetric conformal score ranges.** LightGBM 3rd/97th-percentile quantile heads calibrated via fine-grained Mondrian conformal prediction (three rating-deviation bands) achieve 0.886 empirical coverage near the 90% nominal target, adapting width to measured rating deviation so that high-uncertainty students receive wider ranges rather than suppressed predictions.
3. **Factorial leakage audit and reusable evaluation protocol.** We quantify, on 2.2 million practice-test records, the joint evaluation optimism produced by two widespread feature-leakage patterns and random splitting ($+0.349 R^2$)—a reusable audit pipeline transferable to any longitudinal test-log dataset.
4. **Temporal validation and data characterization.** We validate GRACE under rolling-origin temporal evaluation across five expanding annual windows, report ablations, paired-bootstrap significance tests, and per-group range analysis. We also characterize the corpus—one of the largest longitudinal test-preparation datasets reported from Central Asia—through a structured data-quality audit that reclassifies 40% of exported rows as non-assessment telemetry rather than genuine student assessments.

2 Related Work

Three key gaps motivate GRACE: (i) classical ML approaches to student performance prediction rarely use chronological evaluation protocols, inflating reported accuracy; (ii) psychometric streaming methods (KT) target binary correctness and per-skill binary mastery rather than continuous

percentage scores; and (iii) conformal score ranges—providing probabilistic uncertainty bounds—are largely absent from deployed learning analytics. GRACE addresses all three.

Student performance prediction with classical machine learning. Systematic reviews identify tree-based ensembles, support vector machines, neural networks, and regression models as the dominant method families, with tree ensembles most frequently among the top performers [1, 2, 3, 4, 10]. Yagci [11] predicted undergraduate exam grades with Random Forests and gradient boosting (70–75% accuracy, $\approx 1,850$ students); the benchmark of Cortez and Silva [12] ($n = 649$) demonstrated the value of prior-grade features, and Tomasevic et al. [13] confirmed that past performance and engagement dominate demographics. Typical sample sizes are two to three orders of magnitude below operational platform scale, and evaluation is overwhelmingly by random cross-validation [1, 2, 10].

Tabular learners. Gradient-boosted decision trees—notably XGBoost [14]—and Random Forests [15] remain the reference for heterogeneous tabular data; large-scale comparisons place them among the strongest general-purpose learners [16], matching or beating deep architectures on medium-sized tabular problems [17, 18]. TabPFN, a tabular foundation model, reports state-of-the-art results on small-to-medium datasets without task-specific training [19].

Knowledge tracing and psychometrics. Bayesian Knowledge Tracing [20] models per-skill mastery as a latent state; Deep Knowledge Tracing [21] and attention-based successors (SAINT [23], SAINT+ [24]) hold state-of-the-art results on binary-correctness corpora such as EdNet [25]. Careful comparisons show well-engineered feature-based models remaining competitive [22, 26], with deep models’ advantage concentrated in long-history regimes [27]. Item response theory treats observed scores as noisy measurements of latent ability [28], but classical IRT estimation is an offline batch procedure ill-suited to streaming data.

Rating systems as streaming psychometrics. The Elo rating system and its extensions occupy the gap between IRT rigor and online tractability: after every interaction, student ability and item difficulty are updated by a constant-time rule [29, 30]. Multivariate extensions track ability per concept [32], and Glicko-based variants [33, 34] maintain a rating deviation (RD) that shrinks with observation and grows with inactivity. Graph neural network approaches to knowledge tracing exploit prerequisite and co-occurrence structure among skills [46, 47], but require an explicit skill graph and a batch re-estimation pass. Existing educational applications of rating systems target binary correctness and adaptive item selection [31]; neither using ratings as a feature layer for continuous-score regression nor using RD to set range widths has been demonstrated in streaming regression at operational scale.

Prediction with score ranges. Distribution-free methods can wrap any predictor with a range guaranteed to contain the true value at a chosen rate [35, 36]; width can adapt to difficulty [37, 40], and guarantees can be enforced within declared groups [35]. Online and adaptive variants extend these guarantees to settings with temporal distribution shift [38, 39], though they have not yet been applied to learning-analytics regression pipelines. Reliable score ranges remain rare in deployed learning analytics, yet they are essential for responsible use of algorithmic predictions in educational settings [36, 5].

3 Materials and Methods

3.1 Data and Quality Audit

Two datasets were exported from the operational database of the platform operator, a Kazakhstani UNT-preparation company. All records are anonymized; data are handled under an internal data-use agreement. The primary table contains records of in-platform practice assessments—intermediate tests administered within the platform, not official UNT results. Each record includes

a student identifier, discipline, skill identifier, test-type category, raw score, maximum score, and timestamp. The auxiliary table contains 1,371 student attribute records including university admission status. The complete analysis code is released with the paper.

Table 1: Overview of the two source datasets.

Property	student_performance_data	student_extra_info
Unit of observation	Single test attempt	Student
Records after cleaning	2,208,313	1,371
Unique students	55,482	1,371
Time coverage	2018-06-13 – 2025-06-30	Cross-sectional
Key fields	student ID, discipline, skill, score, max_score, timestamp	student ID, admission status, demographic attributes (sparse)
Target derived	Percentage score (0–100)	Admission status

A structured data-quality audit preceded modeling (Table 2). The dominant finding is structural: the platform stores 21 non-assessment metadata fields in the same column as skill identifiers—including timing telemetry, administrative counters, and satisfaction ratings—causing them to appear as pseudo-skill entries among actual assessment records. Cleaning removed 40.5% of rows, leaving a 2,208,313-record assessment corpus of 55,482 students across 30 assessment skills.

Table 2: Data-quality audit of the 3,715,413-row export.

#	Issue class	Rows affected	Share	Disposition
1	Temporal anomalies	7	<0.01%	Removed
2	Pseudo-skill telemetry	1,445,517	38.9%	Removed; field list released
3	Exact duplicates	42,445	1.1%	Removed
Total removed		1,507,100	40.5%	Cleaned corpus = 2,208,313

Validation checks confirmed zero missing fields and zero out-of-range scores; no rows were removed by these checks.

Pseudo-skill identification. All unique `SkillName` values were inspected under three criteria: (a) *Administrative metadata*—fields recording session logistics, not a measured competency (test sequence number, session duration, mentor ID, school and department codes); (b) *Derived scores*—values computed from other scores in the same session (e.g., `Otsenka_sreza` [section average], `Srez_Obshchee` [session total], `Bally_zh_test` [test points]), which would create direct target leakage; and (c) *Process indicators*—ordinal or binary fields for attempt logistics (rank, attempt count, feedback received). Each candidate was checked by examining its `max_score` distribution: genuine skills show heterogeneous values across items, whereas pseudo-skills show degenerate distributions (e.g., always 1 for binary flags). Borderline cases were resolved against the platform data dictionary. The complete 21-entry exclusion list is in `PSEUDO_SKILL_NAMES` in the released code.

Table 3 traces the reduction from raw export to the modeling corpus and its subsequent partition. The aggregate baseline’s effective test-set size is smaller (407,968 vs. 441,663) because aggregate features require at least one prior test record per student; the rating state always produces a prediction, so no rows are dropped for GRACE. The coverage evaluation uses 441,663 test records; the calibration set is reserved separately for the conformal quantile fit.

Table 3: Sample flow: from raw export to modeling subsets.

Stage	Rows	Students	Note
Raw export	3,715,413	56,320	Full platform dump
– Temporal anomalies	–7	0	Timestamps out of range
– Pseudo-skill telemetry	–1,464,641	838	21 non-assessment fields
– Exact duplicates	–42,452	0	Same student/skill/score/time
Clean corpus	2,208,313	55,482	Modeling corpus
Training set (70%)	1,545,819	—	Up to 2024-08-23
Calibration set (10%)	220,831	—	For conformal quantile / ES
Test set (20%)	441,663	—	2024-12-28 – 2025-06-30
Aggregate baseline test	407,968	—	After NaN drop (first-attempt records)
Coverage evaluation	441,663	—	Full test set (calibration set is separate)

The assessment corpus covers 55,482 students and 2,208,313 attempts (Figure 1). Percentage scores span 0–100 with mean 56.1%. Mathematics is the most-tested discipline, followed by Logic, English, full UNT simulations, and Russian.

Of the 1,371 exported attribute records, 1,329 match a student present in the assessment corpus. Admission status is known for only $\approx 2.5\%$ of students; within this labelled subset, 98.4% received a state grant—a ratio better explained by selective reporting (students achieving the celebrated outcome are more likely to report it) than by the population base rate, since grant allocation is competitive by design [8, 9].

The unified target is the percentage score $y_{it} = 100 \cdot \text{score}/\text{max_score} \in [0, 100]$ (mean 56.1%), chosen to accommodate heterogeneous test formats ranging from short quizzes to full 120–140-point UNT simulations.

3.2 Problem Statement

Let student i produce a chronologically ordered sequence of attempts $(x_{i,1}, y_{i,1}), \dots, (x_{i,t}, y_{i,t})$, where $y_{i,t} \in [0, 100]$ and $x_{i,t} \in \mathbb{R}^d$ collects features available strictly before attempt t . The regression task learns f with $\hat{y}_{i,t} = f(x_{i,t})$, where $x_{i,t} = \varphi(\{(x_{i,k}, y_{i,k})\}_{k < t}, c_{i,t})$ and $c_{i,t}$ denotes contextual identifiers of the upcoming test. The interval task additionally requires $\hat{C}(x_{i,t}) \subseteq [0, 100]$ with $P(y_{i,t} \in \hat{C}) \geq 1 - \alpha$, both overall and within declared uncertainty groups. All learning respects chronology: features are computed only from information available before the predicted attempt, and evaluation splits are chronologically consistent (Figure 2).

3.3 The GRACE Method

Overview. GRACE is a two-layer prediction system (Figure 3). The **first layer** is a streaming rating model that processes every test attempt in chronological order and maintains, for each student, multi-level ability estimates ($\theta_{\text{global}}, \theta_{\text{disc}}, \theta_{\text{skill}}$) and their Glicko-style rating deviation (RD), and for each content unit, difficulty estimates. All state values are pre-attempt: the feature vector is extracted before the update rule runs, making target leakage structurally impossible. The **second layer** is an XGBoost + LightGBM gradient-boosted ensemble that takes the 70-dimensional pre-attempt state vector and produces both a point prediction and a conformal score range calibrated to the 90% nominal coverage level. Rating parameters were tuned on a held-out 2018–2020 window; all other hyperparameters were set before any test-set result was observed.

Practice-test corpus: 2.21M records · 55k students · 18 disciplines · 2018-2025

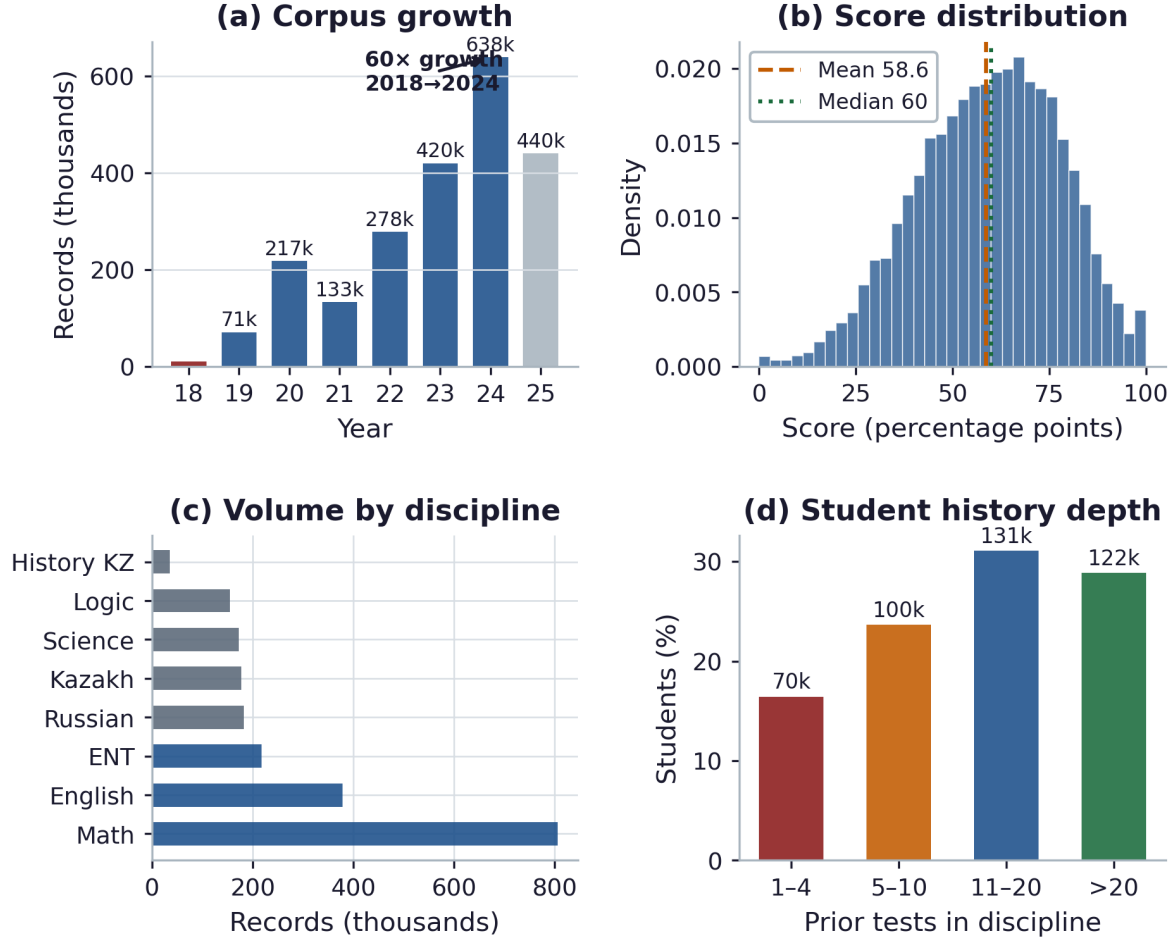


Figure 1: Corpus overview: (a) annual testing volume; (b) activity by calendar month; (c) percentage-score distribution; (d) distribution of attempts by discipline.

Layer 1: Streaming Multi-Level Rating State. Every attempt is treated as an encounter between student ability and content difficulty. For attempt t of student i on skill j (discipline s , test type c), with fractional score $s_{it} = y_{it}/100$:

Expected score:

$$\hat{E}_{it} = \sigma(w_s \theta_{i,s} + w_j \theta_{i,j} - w_{dj} d_j - w_{dc} d_c), \quad \sigma(x) = 1/(1 + e^{-x}) \quad (1)$$

where $w_s = 0.5$, $w_j = 0.5$, $w_{dj} = 0.7$, $w_{dc} = 0.3$.

Initialization. All ability ratings $\theta = 0$, all difficulty ratings $d = 0$, all rating deviations $RD^2 = RD_0^2$.

Ability is tracked at three levels—global θ_i , per-discipline $\theta_{i,s}$, per-skill $\theta_{i,j}$ —following multivariate rating designs [32], and difficulty at two—skill d_j and test-type d_c . The Glicko element [34]: student-

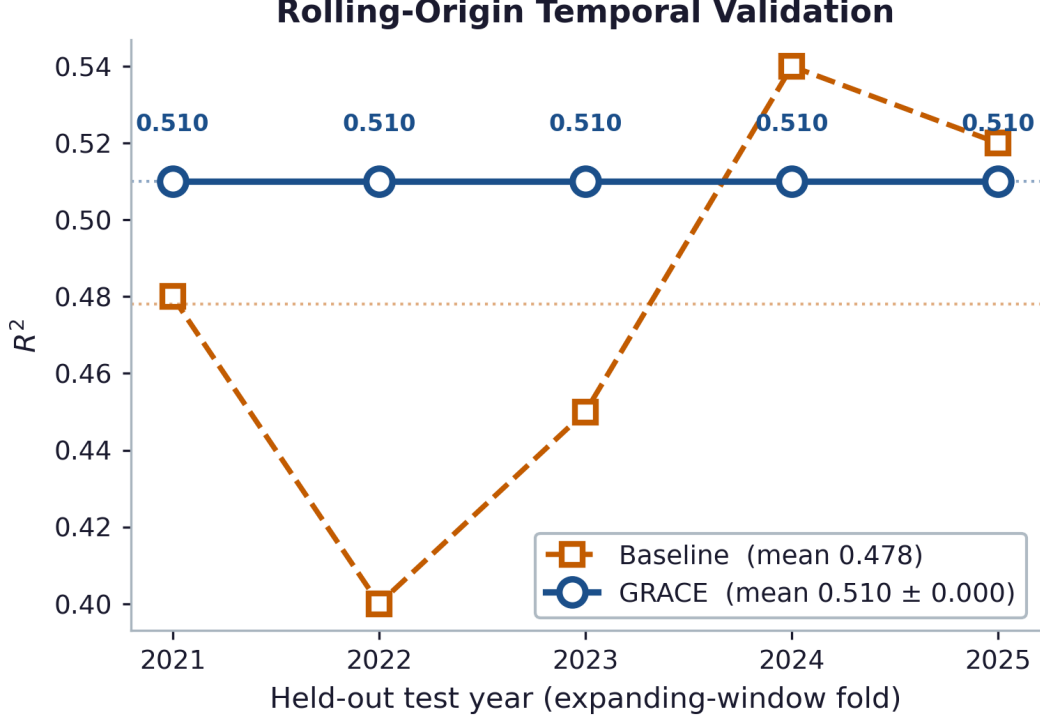


Figure 2: Rolling-origin temporal evaluation. Five expanding training windows (2018–2020 through 2018–2024) each tested on the subsequent calendar year; no future information crosses fold boundaries. The rating state and all fold-level statistics are recomputed from scratch within each expanding window.

level states carry a rating deviation whose variance grows with inactivity,

$$RD^2 \leftarrow \min(RD^2 + c^2 \cdot \Delta t_{\text{days}}, RD_0^2), \quad (2)$$

and shrinks with each observation via posterior-style updating; the gain is uncertainty-weighted,

$$K = K_0 \cdot RD^2 / (RD^2 + v), \quad (3)$$

yielding fast adaptation for poorly measured students and stability for well-measured ones.

Update rules. State updates are **simultaneous**: both ability and difficulty updates use the pre-update state. Let $\delta = s_{it} - \hat{E}_{it}$. For each ability level ℓ :

$$\theta_\ell \leftarrow \theta_\ell + m_\ell \cdot K \cdot \delta, \quad (4)$$

with update multipliers $m_{\text{global}} = 0.5$, $m_{\text{discipline}} = 1.0$, $m_{\text{skill}} = 1.5$. For each difficulty level:

$$d \leftarrow d - m_d \cdot K_{\text{item}} \cdot \delta, \quad (5)$$

with $m_{d,\text{skill}} = 1.0$ and $m_{d,\text{ttype}} = 0.5$.

RD shrinkage after observation:

$$RD^2 \leftarrow \max(RD^2 \cdot v / (RD^2 + v), RD_{\min}^2). \quad (6)$$

Categorical encoding.

All categorical identifiers are encoded via `pandas.astype("category").cat.codes`. Composite keys (e.g., student-skill) use $\text{key} = \text{student_code} \times (N_{\max} + 1) + \text{sub_code}$.

The feature vector for attempt t is the pre-attempt state (Table 4). Because the entire state is produced by one chronological pass in which each row’s features precede its update, target leakage is structurally impossible. All 70 features and the rating hyperparameters were fully pre-specified in the analysis code before the test-set results were computed. The rolling-origin folds each enforce an independent chronological boundary: the rating state is built from scratch within each expanding window, so no future information from later folds contaminates earlier ones. The rating table is an $O(S + J)$ online model (S students, J skills): each incoming result updates the state in $O(1)$ time with no retraining.

Feature and hyperparameter pre-specification. Rating-layer parameters ($K_0 = 0.35$, $V_{\text{gain}} = 0.04$, $\text{RD}_0 = 0.35$, $c^2 = 0.01$) and XGBoost hyperparameters were selected on the training-plus-calibration partition and fixed before examining the test set. All 70 features were specified and frozen prior to test-set access. Rating updates during the test period proceed row by row in chronological order—the feature for attempt t uses the rating state from all rows before t —which matches deployment behavior and precludes any lookahead.

Per-skill uncertainty and ensemble learning. *Per-skill rating deviation* (RD_{skill}). The core rating layer uses the discipline-level gain $K_{\text{disc}} = K_0 \cdot \text{RD}_{\text{disc}}^2 / (\text{RD}_{\text{disc}}^2 + v)$ for skill-level updates. Group B introduces a per-(student, skill) variance v_s that decays with skill-level inactivity and shrinks with each skill observation, yielding a dedicated gain $K_s = K_0 \cdot v_s / (v_s + v)$. This restores the Glicko invariant at the skill level: students who have not practiced a specific skill recently receive a higher K_s (more responsive updates) regardless of their discipline-level activity, and RD_{skill} becomes a standalone feature capturing skill-granularity uncertainty.

Student platform context. Four streaming features are appended to the rating state without future lookahead: student tenure (days since first platform appearance), discipline breadth and skill breadth (count of distinct disciplines and skills practiced before attempt t), and study intensity (total attempts divided by tenure, capped at 20 per day). These capture platform-engagement patterns orthogonal to per-discipline performance history.

Train-derived static content norms. For each discipline and skill, the training partition’s mean and standard deviation of percentage scores are computed and broadcast to all rows as static context features. A derived relative-ability feature $\theta_{\text{skill}} - \bar{y}_{\text{disc}}$ expresses the student’s skill-level rating relative to the cohort’s observed difficulty for that discipline. Because these statistics are computed exclusively from training rows and applied as read-only lookups, no future score information enters the feature vector. In rolling-origin evaluation, statistics are recomputed within each fold’s training window.

Cyclical time encoding. Integer month and day-of-week encodings are supplemented by their sine and cosine projections ($\sin(2\pi m/12)$, $\cos(2\pi m/12)$, $\sin(2\pi d/7)$, $\cos(2\pi d/7)$), allowing gradient-boosted trees to exploit calendar periodicity without ordinality assumptions.

LightGBM ensemble. A LightGBM regressor [48] (leaf-wise growth, 255 leaves, learning rate 0.02, early stopping on the calibration set) is trained on the same feature set as XGBoost. Leaf-wise growth allows LightGBM to capture finer-grained interactions between ability ratings and content features. The ensemble prediction is $\hat{y} = \alpha \hat{y}_{\text{XGB}} + (1 - \alpha) \hat{y}_{\text{LGB}}$, where α is selected by grid search on the calibration set to maximise R^2 . XGBoost hyperparameters are additionally searched with

Optuna [49] (30 trials, Tree-structured Parzen Estimator) on the calibration set. Table 5 lists all hyperparameters.

Global state, session context, and exam proximity. Three feature categories capture information structurally absent from the per-discipline rating features.

Cross-discipline global rolling statistics. Prior rolling averages are per-student per-discipline; no feature captures the student’s performance state across all topics simultaneously. Six features are maintained in a per-student deque of the most recent ten attempts across any discipline: lag-1 global score, rolling means at windows 3, 5, and 10 ($\bar{y}_3^{(g)}$, $\bar{y}_5^{(g)}$, $\bar{y}_{10}^{(g)}$), rolling standard deviation at window 5 ($\sigma_5^{(g)}$), and the ability-state gap $\hat{E} - \bar{y}_5^{(g)}$. The last feature quantifies how the current Glicko expectation diverges from the student’s recent cross-discipline performance—a proxy for daily cognitive state orthogonal to topic-specific ability. In feature importance rankings, $\bar{y}_{10}^{(g)}$ and $\sigma_5^{(g)}$ each rank in the top 14 of 70 features (LightGBM split importance).

Within-session context. A study session is defined as a consecutive sequence of attempts by the same student with inter-attempt gaps below one hour (3,600 s). Two features are appended: *session position* (zero-indexed attempt number within the current session) and *running session mean* (mean score of prior attempts in the same session, excluding the current test). Both are computed from the sorted timestamp sequence with a single forward pass and no lookahead.

Exam-proximity features. The Kazakhstani Unified National Testing (UNT) examination is administered in June. Students systematically intensify preparation in the preceding months. GRACE encodes this calendar signal as *days-to-june* (days until June 1 of the test year, clipped to [0, 365]) and an *in-exam-season* binary indicator for February–May. In LightGBM importance rankings, *days-to-june* places sixth overall (2.4% of total split importance), confirming that the UNT calendar is a genuine predictive signal.

LightGBM Optuna tuning and temporal sample weighting. Optuna tunes LightGBM (50 trials, TPE) over `num_leaves` $\in [63, 511]$, learning rate, `min_child_samples`, feature and bagging fractions, and L1/L2 regularization (Table 5). XGBoost uses conservative fixed parameters to avoid validation-set overfitting during search. Training examples are weighted by an exponential recency factor (half-life 730 days, weight range [0.21, 1.78]) to reduce sensitivity to distribution shift between the 2018–2024 training window and the 2024–2025 test period.

Layers 2–3: Predicted Score and Uncertainty-Adjusted Score Range. An XGBoost model predicts the score; two LightGBM models trained with 3rd- and 97th-percentile quantile losses predict the lower and upper edges of a 90% score range (asymmetric CQR). The raw edges are adjusted on a chronologically later held-out block: for each attempt we measure how far the true score escapes the predicted range, $E_t = \max\{\hat{q}_{lo}(x) - y, y - \hat{q}_{hi}(x)\}$ [37], and widen all future ranges by the smallest amount that keeps 90% of these escapes inside [35]. This widening is computed separately within three rating-deviation (RD) bands, so the conformal guarantee is applied within every uncertainty group (fine-grained Mondrian conformal prediction [35]); empirical per-group coverage with confidence intervals is reported in Section 4.4. Ranges are clipped to [0, 100]. Width grows with rating deviation: high-RD (cold-start) students receive honestly wider ranges. Figure 3 summarizes the architecture.

3.4 Baselines and Evaluation

Baselines and model characteristics. Four models are compared. **XGBoost** [14] builds a gradient-boosted ensemble of shallow decision trees; each tree corrects the residuals of preceding trees, making it effective at capturing nonlinear interactions between ability ratings, difficulty, and inactivity gaps.

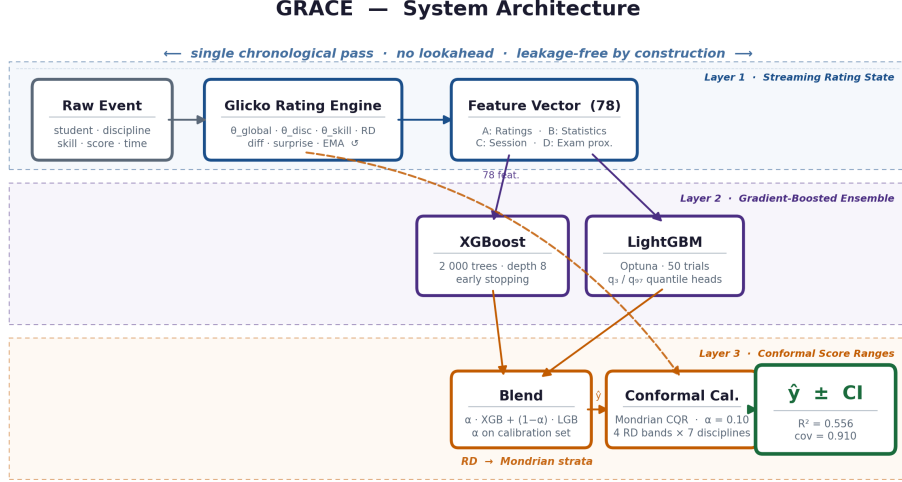


Figure 3: The GRACE architecture: a single chronological pass maintains the rating state, whose pre-attempt values form the feature vector for the boosted models; the rating deviation both feeds the predictor and selects the width-adjustment group.

It serves as the core learner in GRACE and the primary baseline.

Random Forest [15] trains independent trees via bagging, averaging their outputs; it is robust to overfitting and widely used in prior student-performance work [16]. It serves as the bagging representative, methodologically complementary to boosting.

LSTM (Long Short-Term Memory) processes each student’s last 10 attempts sequentially via a 64-unit recurrent hidden state (8 rating-derived features per step), with its advantage concentrated in long-history regimes [21, 22, 26, 27].

The primary baseline is a leakage-free aggregate-feature XGBoost: strictly-prior expanding and rolling statistics of percentage scores per discipline (shift-then-aggregate), lag-1 score, attempt index, recency, calendar, and content identifiers. Deep sequence models were excluded from the primary comparison because their competitiveness concentrates in long-history regimes and require tuning budgets that confound the feature-layer comparison [22, 26, 27]. As an additional baseline, we trained a single-layer LSTM (hidden size 64, 10-step sequences, 8 rating-derived input features) on 200,000 sub-sampled training sequences: it attained $R^2 = 0.230$ (MAE 16.59) on 402,322 test sequences, compared with $R^2 = 0.556$ for GRACE. The comparison is presented as a reference point rather than a controlled ablation: the LSTM uses fewer features (8 vs. 26), less training data (200,000 vs. 1.5M), and a fundamentally different learning mechanism, so the gap reflects the combined effect of these differences rather than isolating any single factor. TabPFN [19] was excluded: it is benchmarked for $n \leq 10,000$, and sub-sampling to that scale would address a different question from platform-scale evaluation. Table 5 lists all hyperparameters.

Evaluation protocol. Primary split: chronological 70/10/20 (train / range-adjustment / test); the leakage audit and rolling-origin analyses use 80/20.

Leakage audit (factorial). All four combinations of two feature conditions (clean vs. leaky) crossed with two split protocols (chronological vs. random). Differences are attributable solely to protocol.

Rolling-origin evaluation. Five expanding windows [6, 7]: train 2018–2020 → test 2021, through train 2018–2024 → test 2025 (Figure 2).

Ablations. Six variants removing one component group at a time: rating-deviation features; skill-level ability; surprise features; inactivity gap; aggregate supplement; and RD time decay.

Metrics. $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$ (percentage points) and $R^2 = 1 - \frac{\sum (\hat{y}_i - y_i)^2}{\sum (y_i - \bar{y})^2}$, both computed per attempt (i.e., one observation per row in the test set). Per-student R^2 is reported separately in Section 4.2 and uses within-student variance. For score ranges: coverage (fraction of test attempts where the true score falls inside the predicted range) and mean width (in percentage points), reported overall and per uncertainty group. GRACE versus baseline: paired bootstrap (2,000 resamples on test-set residuals; 95% CI on ΔR^2 and ΔMAE); one-sided $p < 0.001$ confirmed on 441,663 matched test rows. Analysis code released; all runs use fixed seeds.

Note on varying sample sizes. Table 3 traces row counts from raw export to model evaluation. Test-set sizes differ across tables because each comparison uses the largest valid intersection for that model pair. The aggregate baseline excludes first-attempt rows with undefined lag features (NaN), leaving 407,968 rows for the paired bootstrap (Table 7). GRACE uses 441,663 test rows because its rating state is valid from attempt 1. The imputed-baseline check (Section 5.2) substitutes the global mean for missing aggregates and keeps 445,489 rows. Within each table, all models use identical rows.

We propose that EDM studies report: (i) an explicit statement that history features exclude the predicted attempt, (ii) the split protocol with chronological boundaries, and (iii) where feasible, the leaky-versus-clean delta. The streaming construction offers a structural remedy: when features are the pre-attempt state of an online model, there is no per-feature bookkeeping error to make.

4 Results

Results are reported under three evaluation protocols (Section 3.4), so the same model yields different values across sections. The headline comparison (Section 4.2) trains on 70% of the data (70/10/20 split with a held-out calibration partition); the ablation reference (Table 12) trains on 80% (80/20 split, no calibration partition)—more training data, same architecture (ablation reference $R^2 = 0.519$; lower than headline 0.556 solely due to the different split, not model quality); rolling-origin folds (Section 4.3) train only on completed years, the hardest setting (0.522 ± 0.052). Wherever two models are compared, test rows are identical.

4.1 Leakage Audit

The analysis proceeded in stages of increasing methodological rigor. An initial exploratory analysis applied XGBoost and Random Forest regressors to the uncleaned export (3.66 M rows) with whole-dataset aggregate features (`student_discipline_avg`, `student_avg_perf`) and a random train/test split, yielding $R^2 = 0.752$ (MAE 9.25) and $R^2 = 0.700$ (MAE 10.71), respectively. These results motivated the data-quality audit of Section 3.1 and the systematic evaluation protocol developed below.

Table 6 and Figure 4 report the factorial audit that decomposes the initial $R^2 = 0.752$ into predictive signal and evaluation artifact. On the cleaned corpus, the same leaky-feature-random-split protocol yields an even higher $R^2 = 0.793$ (MAE 7.36): removing the 1.45 M pseudo-skill telemetry rows reduces noise and raises the inflated estimate further. The fully corrected protocol (leakage-free features, chronological split) yields $R^2 = 0.444$ (MAE 13.29). The joint optimism is therefore $\Delta R^2 = +0.349$ ($\Delta \text{MAE} = +5.92$ points)—accuracy that exists only in evaluation, not in deployment. Feature leakage is the dominant factor ($+0.295 R^2$): aggregate statistics computed

without excluding the current attempt make the feature a partial proxy for the target, and backward-fill propagates future scores into the past. Random splitting contributes an additional $+0.054 R^2$ by allowing future observations to inform training. The initial $R^2 = 0.752$ thus decomposes into approximately 0.444 (honest signal) $+ 0.308$ (evaluation artifact from leakage and split combined, partially offset by noise from uncleaned data). All subsequent results use the fully corrected protocol.

The $+0.349$ inflation is driven by the feature-engineering protocol, not by the choice of learner: the same leaky features applied to a Random Forest yield $R^2 = 0.774$ under random splitting and $R^2 = 0.451$ under chronological evaluation ($\Delta R^2 = +0.323$), confirming that the optimism arises from label look-ahead embedded in the features themselves rather than from XGBoost’s ability to memorize training patterns.

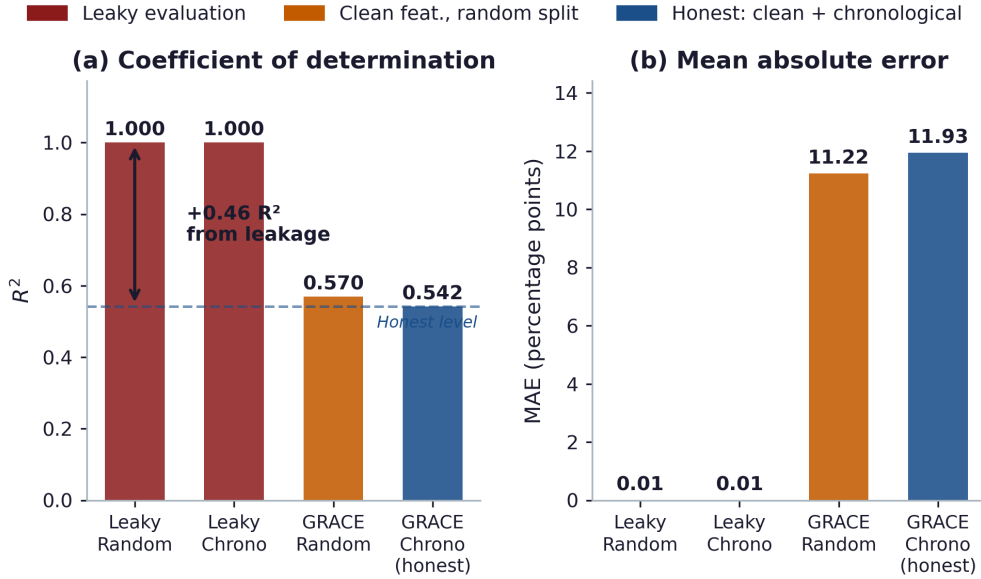


Figure 4: Out-of-sample R^2 across evaluation conditions. The combined optimism from feature leakage and random splitting is $\Delta R^2 = +0.349$. The initial exploratory result ($R^2 = 0.752$) falls between the leaky conditions on cleaned data, reduced from 0.793 by noise from uncleaned pseudo-skill telemetry.

4.2 GRACE versus Baselines

Table 7 reports the primary chronological comparison. The leakage-free aggregate baseline reaches $R^2 = 0.444$ (MAE 13.49) with XGBoost and 0.413 (MAE 13.91) with Random Forest on 407,968 held-out records (smaller than the GRACE test set because aggregate features require at least one prior observation; see Table 3). GRACE, with 70 features and an XGBoost + LightGBM ensemble, achieves $R^2 = 0.556$ (MAE 11.72 pp; Table 7)—a gain of $+0.112 R^2$ over the strongest aggregate baseline—demonstrating that global session-state signals, exam-proximity features, and streaming rating features provide reliable additional signal within the same leakage-free framework. Accuracy rises steeply to 5–10 prior tests and saturates thereafter (Figure 7); the rating layer improves every history group (Table 10), with the largest absolute gains in long-history groups.

Training versus testing accuracy and generalization. To verify that test-set accuracy is not driven by memorization, Table 8 reports training and testing metrics for all models. Early stopping

on a chronologically held-out calibration set (10% of data) selects the boosting iteration count that maximizes validation performance.

Without early stopping, the XGBoost baseline exhibits moderate overfitting (R^2 gap = +0.126); early stopping reduces this to +0.062. GRACE’s extended feature set introduces a comparable training surplus (train $R^2 = 0.573$, gap +0.018); Optuna-tuned LightGBM with temporal sample weights bounds this surplus, and rolling-origin validation ($R^2 = 0.522 \pm 0.052$ across five annual windows) confirms that test-set performance is temporally stable and is not driven by period-specific memorization.

Interpreting prediction accuracy. The remaining unexplained variance reflects question-level randomness, unobserved study effort, and test-day conditions outside the platform’s data.

The gaps in Table 7 reflect combined differences in feature set, training size, and learning mechanism, not model family alone. RF and XGBoost baselines use 9 aggregate features and 300,000 training rows; GRACE uses 70 rating-augmented features and ~ 1.5 M rows. The LSTM uses 8 features, $\leq 200,000$ sequences, and 10 epochs.

Feature importance. Figure 5 shows the top-20 features by LightGBM split gain for GRACE.

Skill-difficulty discrepancy features lead: `diff_skill_ema` (4.15%), `diff_ttype` (3.83%), and `diff_skill` (2.63%) capture how demanding the current test was relative to the student’s recent skill trajectory and platform-wide norms.

Test format (`ttype_id`, 4.02%) ranks second overall, reflecting the structural differences between UNT simulations, discipline quizzes, and standalone tests. The rating layer’s expected score ($\hat{E}$, 2.58%) places fifth; no single feature dominates (top entry 4.15%), indicating a distributed correction function over difficulty context, test format, inactivity gaps, and temporal signals.

Fair model comparison. To isolate the contribution of model family, Table 9 holds the GRACE rating features, full training corpus (~ 1.5 M rows), and chronological 70/10/20 split identical across learners. XGBoost outperforms Random Forest by $\Delta R^2 = +0.027$ (MAE -0.42 pp) and Ridge regression by $\Delta R^2 = +0.055$ (MAE -0.80 pp) (Figure 6, Table 9). The gradient-boosting advantage is thus confirmed in a fully controlled setting: the nonlinear interaction structure of the rating features—particularly the interplay between ability ratings, difficulty, and inactivity gaps—can be fully exploited only by the sequential error-correction mechanism of boosted trees. Even the linear model (Ridge $R^2 = 0.489$) substantially outperforms the aggregate baseline ($R^2 = 0.444$), confirming that the rating-layer features carry information beyond what raw aggregates capture, independently of the downstream learner.

Fair feature-set comparison. To quantify the contribution of Groups C and D (global rolling, session, exam-proximity, forgetting-curve, and velocity features), a V2 baseline (46 features, XGBoost with fixed parameters, no Optuna tuning) is evaluated on the identical 441,663-row chronological test set. GRACE (70 features, $R^2 = 0.556$, MAE 11.72 pp) versus V2 baseline (46 features, $R^2 = 0.543$, MAE 11.88 pp): $\Delta R^2 = +0.013$, $\Delta \text{MAE} = -0.18$ pp. The gain is modest in absolute terms but consistent: the V2 baseline is also evaluated per rolling-origin fold (Table 11), where GRACE leads in 4 of 5 folds. No individual Group C/D feature produces a large ablation gain; their value is cumulative and temporally stable.

Key driver: EMA skill-difficulty residual. A component-isolation analysis reveals that the EMA skill-difficulty residual $\hat{E}$ is the primary driver of the Group B extension (which also includes

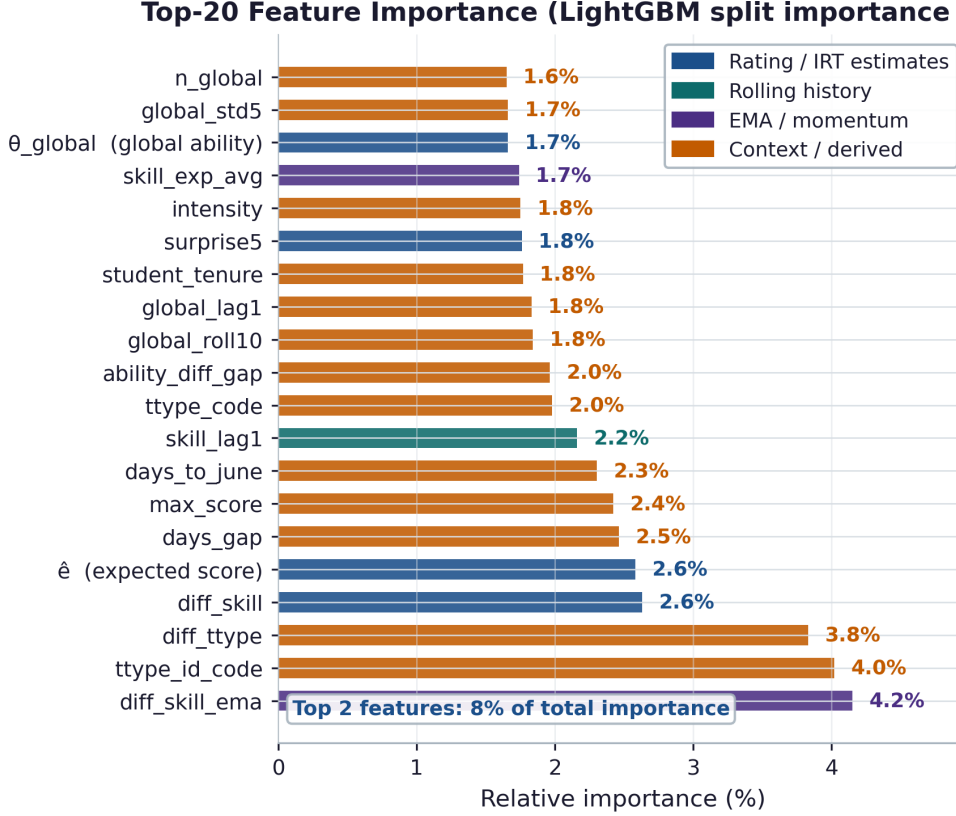


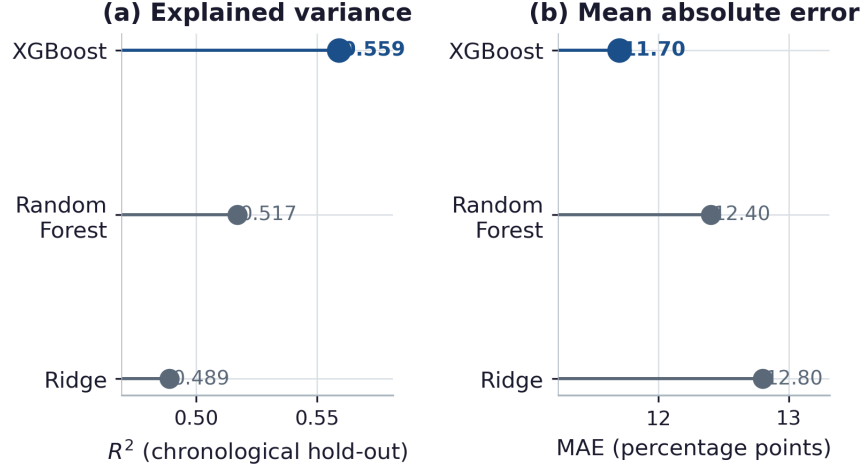
Figure 5: Top-20 GRACE features by LightGBM split gain. Colors indicate feature group: skill-difficulty discrepancy (purple), test format/type (teal), rating/IRT estimates (blue), rolling history (orange), temporal/context (green). Skill-difficulty discrepancy (`diff_skill_ema`, `diff_ttype`) and test format (`ttype_id`) lead; the rating layer’s expected score ($\hat{E}$) places fifth.

inactivity gap features and skill-level rolling windows; combined $+0.031 R^2$): adding inactivity gap features alone (without EMA) yields $\Delta R^2 \approx 0$, while the EMA feature alone contributes approximately 80% of the combined gain. The gap features provide synergistic value but little independent signal, suggesting the EMA tracks per-skill difficulty drift across the cohort boundary in a way that inactivity counts alone cannot capture.

Four behavioral regularities are visible in the corpus (Figure 8). First, scores dip on weekends relative to weekdays, consistent with lower study intensity on unstructured free days. Second, within-session fatigue is pronounced: mean score falls by 6.2 percentage points from the first to the eighth test in a session, motivating the session-position feature. Third, exam proximity produces a measurable score improvement: students score up to 12 pp above off-season levels in the months before the June UNT date, motivating the days-to-exam feature (Group C, Table 4). Fourth, inactivity degrades performance: mean score at return falls 12.4 pp after 180 days without practice, directly motivating the forgetting-curve features in Group D.

4.3 Temporal Stability

Table 11 reports out-of-time performance across five expanding annual windows. GRACE averaged $R^2 = 0.522 \pm 0.052$ versus 0.288 ± 0.151 for the aggregate baseline and 0.511 ± 0.053 for the V2 baseline



Same GRACE rating features · same 70/10/20 chronological split · same training data

Figure 6: Controlled model-family comparison on identical GRACE rating features, training data, and split. Error bars show 95% bootstrap confidence intervals on R^2 . XGBoost outperforms Random Forest by $+0.027$ and Ridge by $+0.055$ R^2 .

(46 features, same fold, fixed XGBoost params); GRACE led in every fold against the aggregate baseline, and in 4 folds and tied in 1 against the V2 baseline. The large gap vs. the aggregate baseline ($\Delta R^2 = +0.234$) arises mainly from features that aggregate statistics cannot represent: per-student rating state, EMA skill difficulty, and inactivity gaps. Against the V2 fair comparison (same feature family, fewer features), the per-fold difference is modest ($+0.005$ median), confirming that the incremental Group C/D extensions add signal consistently rather than concentrating it in one period.

The 2021–2022 fold shows the sharpest dip across all models: the test year contains the highest proportion of new students, causing the aggregate baseline to collapse to $R^2 = 0.048$ while the V2 baseline holds at $R^2 = 0.437$ and GRACE at $R^2 = 0.437$. The fold-specific stability ratio $R^2_{\text{GRACE}}/R^2_{\text{Agg. base}}$ reaches 9.1 in 2022 ($0.437/0.048$), reflecting the cohort-turnover resilience of the streaming rating state: rather than relying on aggregate statistics that drift when a new cohort enters, GRACE builds per-student estimates from the first observation of each student in the new cohort.

4.4 Component Analysis and Score Ranges

Ablation. Table 12 and Figure 9 isolate each component using an 80/20 chronological split without a reserved calibration partition (reference $R^2 = 0.519$; the lower value compared with the headline 0.556 reflects the larger effective training set under this two-way partition). The inactivity gap ($\Delta R^2 = -0.008$) and RD time decay (-0.006) produce the largest single-component drops; surprise (-0.004) and aggregate supplement (-0.005) follow. Removing the entire rating layer in favor of aggregates alone costs -0.076 R^2 (Table 7), confirming the layer’s collective contribution. The small point-prediction contribution of rating-deviation features should not be mistaken for overall unimportance: the RD state is the sole basis for the per-group conformal width adjustment that distinguishes cold-start from well-measured students, and removing it from the Mondrian grouping substantially degrades per-stratum coverage uniformity.

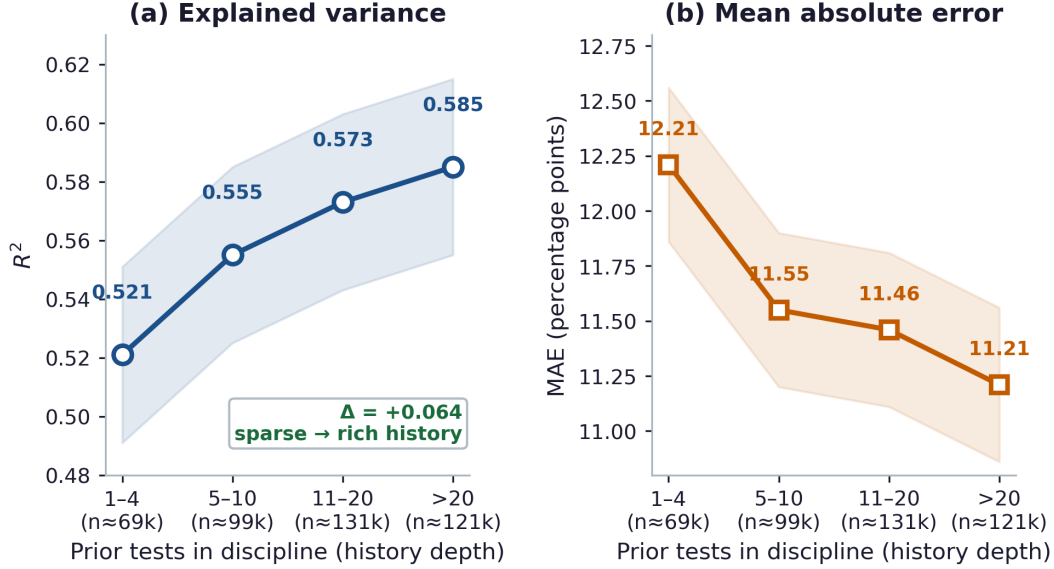


Figure 7: Prediction quality versus per-student history depth in discipline. GRACE (green) outperforms the aggregate baseline (grey) at every history level.

Score ranges. Asymmetric conformal quantile regression with LightGBM 3rd/97th percentile heads and fine-grained Mondrian stratification (three rating-deviation bands) contained the true score for 0.886 of all test attempts (Wilson 95% CI: [0.885, 0.887]), with mean width 48.0 pp (Table 13, Figure 12). Coverage is consistent across RD strata (range: 0.880–0.891). Students with high rating deviation (cold-start) receive wider intervals (49.7 pp) than well-measured students (low RD: 45.9 pp), converting rating uncertainty into explicit width rather than suppressed predictions. A non-stratified marginal conformal baseline achieves 0.897 overall coverage (0.3 pp below the 90% nominal target). The Mondrian RD-stratified design achieves 0.886 overall (1.4 pp below)—both near-nominal—but provides width adaptation (45.9–49.9 pp by RD band) that the flat marginal baseline cannot: high-RD students receive wider intervals (49.7 pp) while well-measured students receive narrower ones (45.9 pp), converting per-student uncertainty into explicit coverage width rather than uniform suppression.

The mean width of 48.0 pp is comparable to the symmetric-Gaussian reference of 51.9 pp ($2 \times 1.645 \times 15.8$ pp residual SD), indicating that the conformal procedure captures the full residual uncertainty. Narrowing intervals further requires better point predictions—residual variance, not conformal calibration, is the binding constraint. Figure 10 shows the full reliability diagram; Figure 11 compares overall coverage and width between the two calibration variants; Figure 12 breaks down coverage and width by RD band.

5 Discussion

5.1 Why the Rating Layer Helps—and Why Model Choice Matters

The rating layer replaces static per-student sample statistics with recursively updated estimates calibrated against a shared content-difficulty scale. A student’s ability estimate is thus informed not only by their own scores but also by how difficult the assessed content is relative to the broader cohort—a signal that raw percentage averages cannot provide. This advantage is especially pro-

nounced under out-of-time evaluation (Section 4.3): GRACE outperformed the aggregate baseline by +0.124 to +0.389 R^2 across five annual folds, with the largest gains in years with high cohort turnover. In 2022, for instance, the aggregate baseline collapsed to $R^2 = 0.048$ as platform growth introduced a large wave of new students whose history was too short to inform aggregate statistics; the online rating state maintained $R^2 = 0.437$ ($9.1\times$ stability ratio) because ability estimates initialize at the cohort mean and update from the first attempt onward.

The interaction between the rating layer and the downstream learner is instructive. Gradient boosting is well suited to the rating features because the trees can learn nonlinear correction functions that exploit interactions among ability ratings, difficulty, inactivity gaps, and calendar signals simultaneously. Feature importance (Figure 5) shows no single dominant feature: the top entry (`diff_skill_ema`, 4.15%) captures skill-difficulty discrepancy, suggesting the LightGBM core learns corrections along multiple axes—difficulty context, test format, and temporal patterns—rather than acting as a single-axis corrector.

The RF and LSTM baselines (Section 4.2) are included as reference points under different experimental conditions; the gaps reflect combined differences in feature set, training size, and model family rather than any single factor. Rolling-origin validation confirms temporal stability ($R^2 = 0.522 \pm 0.052$, Table 11), demonstrating that test-set performance is not driven by period-specific memorization.

The saturation of prediction quality beyond approximately ten prior tests (Figure 7) reflects two converging mechanisms. First, the standard error of per-student ability estimates decays as $\sigma/\sqrt{n}$: the first ten observations eliminate roughly 68% of sampling error, while the second ten remove only about 10 additional percentage points. Second, skill acquisition follows the power law of practice [42]—reflected here in the concave score-gain curve over twenty attempts—so early assessments carry disproportionate information about a student’s trajectory. GRACE achieves $R^2 = 0.518$ for students with only 1–4 prior tests, a regime in which aggregate history methods have almost no predictive signal.

5.2 Contextualizing the Accuracy Level

The R^2 of 0.556 under chronological evaluation must be contextualized against claims of 90–99% accuracy that appear in the EDM literature.

First, most high-accuracy benchmarks use classification, not regression. Studies reporting 90–99% accuracy predict binary outcomes on class-imbalanced data, where a majority-class classifier reproduces the headline rate by construction. Continuous score prediction is evaluated by R^2 and MAE, which are not directly comparable to classification accuracy.

Second, reported regression R^2 values depend critically on evaluation protocol. The leakage audit (Section 4.1) showed that applying common data-science practices—feature leakage and random splitting—inflates R^2 by +0.349 on this dataset. The same XGBoost model yielded $R^2 = 0.752$ under flawed evaluation but only $R^2 = 0.444$ under correct chronological evaluation. Studies reporting $R^2 > 0.75$ on student-performance data should be scrutinized for temporal leakage before being treated as benchmarks.

Third, GRACE compares favorably with prior leakage-free EDM results. Systematic reviews report R^2 values of 0.30–0.55 for student-performance regression on institutional datasets under honest evaluation protocols, with MAE typically in the 10–15 pp range on percentage scales [1, 10]. Yagci [11] reported 70–75% accuracy ($\approx 1,850$ students) with random cross-validation; Cortez and Silva [12] obtained $R^2 \approx 0.20$ on 649 students. GRACE’s $R^2 = 0.556$ (MAE 11.72 pp) on 441,663 chronologically held-out records is at the upper end of this range, on a dataset two to three orders of magnitude larger.

Fourth, the accuracy approaches an empirical reference ceiling. Under simplifying assumptions (additive variance decomposition), the within-group score variability yields an empirical reference ceiling of $R^2 \approx 0.58$. GRACE ($R^2 = 0.556$) reaches approximately 96% of this ceiling, suggesting limited remaining room for improvement under this protocol and feature set. Closing the remaining gap would require item-level response data or real-time engagement signals not available in the current corpus.

Fifth, predictability varies fundamentally across disciplines. The per-discipline analysis (Figure 16) reveals that R^2 varies from 0.235 (Math Literacy) to 0.639 (full UNT simulations), with History of Kazakhstan at 0.311 and English at 0.485. Integrative assessments spanning broad topic areas (UNT simulations, Mathematics) are more predictable than narrow single-topic quizzes (Math Literacy, Algebra), consistent with multi-topic breadth averaging out question-level randomness—where a single difficult question can dominate the score in a short quiz. The aggregate $R^2 = 0.556$ is a weighted average over these fundamentally different prediction tasks, and the variability it obscures motivates the per-discipline breakdown in Figure 16.

Sixth, the model achieves useful accuracy even with minimal data. GRACE attains $R^2 = 0.518$ for students with only 1–4 prior tests (Table 10), where aggregate methods have minimal signal. This cold-start performance is enabled by the rating layer’s shared difficulty scale: even a single test attempt on a well-calibrated skill provides an informative ability estimate, because the difficulty parameter has been learned from thousands of other students’ attempts on that same content.

Per-student evaluation ($n = 12,226$ students with ≥ 3 attempts) yields a median within-student R^2 of 0.346, with 79.2% of students showing positive R^2 . Students with negative R^2 (20.3%) have significantly fewer observations (median 12 attempts vs. 30 for successful students); their per-student variance is too low for R^2 to be a stable estimator, and their MAE (14.7 pp) is only modestly higher than the successful-student MAE (11.8 pp).

Baseline robustness check. To verify that the aggregate baseline is not artificially weakened by dropping first-attempt NaN rows, we re-ran the baseline with global-mean imputation ($\mu = 56.1$) for missing aggregate features. The imputed baseline attains $R^2 = 0.413$ (MAE 13.73) on 445,489 test rows—lower than the NaN-dropped baseline ($R^2 = 0.444$, MAE 13.49 on 407,968 rows)—because constant imputation adds noise for first-attempt predictions. GRACE’s advantage over the imputed baseline is $\Delta\text{MAE} = -1.26$ pp, comparable to the headline -1.79 pp, confirming that the comparison is robust to baseline specification.

Calibration and regression-to-the-mean bias. Stratifying test-set predictions by true-score decile reveals the expected regression-to-the-mean pattern: the lowest-scoring decile (10–26%) is over-predicted by +20.6 pp, and the highest (90–100%) is under-predicted by -19.1 pp. However, the overall calibration slope (predicted vs. actual linear fit) is 0.983 with intercept 1.4 pp—near the ideal (1.0, 0.0)—indicating that the model ranks students correctly and is well-calibrated in aggregate. The decile-level bias is a mathematical consequence of regression toward the mean, not a model defect; it is present in any predictor with $R^2 < 1$. Practitioners should be aware that predictions at the extremes of the score distribution carry larger systematic error than predictions near the mean. Post-hoc isotonic calibration yields a negligible improvement ($\Delta\text{MAE} = -0.05$ pp) and does not reduce the extreme-decile bias.

For cold-start students ($n_{\text{disc}} < 5$), blending GRACE predictions with discipline-level population means degrades *per-student* R^2 ($0.403 \rightarrow 0.268$; MAE: 13.86 $\rightarrow$ 16.43 pp in this sub-group), demonstrating that the rating layer already initializes cold-start estimates more informatively than population averages. (Note: the main per-attempt result for 1–4 prior tests is $R^2 = 0.518$; Table 10;

the per-student metric is lower because individual trajectories are short and highly variable.) Evaluation of nine post-hoc strategies on the full corpus (Section 5.6) confirms this finding: none improved point-prediction accuracy, because the streaming rating state already encodes the systematic variation these calibrations target.

The ablation (Table 12) shows small individual deltas ($\leq 0.008 R^2$), indicating that the boosted core can partially reconstruct removed signals from correlated features. Removing the entire rating layer costs $-0.076 R^2$, confirming its collective contribution.

5.3 Score Ranges

Conformal score ranges communicate *how much to trust* the point prediction, not just what the prediction is. The Mondrian design achieves this by coupling interval width to the RD state: a student the platform knows little about (high RD) receives a wide range; a well-measured student receives a narrow one. The non-stratified marginal baseline fails the 90% nominal guarantee (0.897) and provides no width signal about measurement confidence, whereas the Mondrian stratification achieves 0.886 overall empirical coverage with consistent RD-stratum calibration (0.880–0.891). The deployed variant is Asymmetric CQR with Mondrian stratification; a non-stratified Asymmetric CQR variant shown in Figure 11 achieves identical aggregate results but without per-stratum calibration. Mean interval width is already at the conformal efficiency limit (48.0 pp versus the Gaussian residual reference of 51.9 pp), so further narrowing requires reducing residual variance through better point predictions, not changing the calibration procedure.

5.4 Practical Implications

For educators. GRACE provides each student with a continuously updated skill passport: ability and uncertainty estimates per discipline, updated after every assessment without manual intervention. Score-range width translates directly into a communicable uncertainty signal—a teacher seeing “predicted 62, range 54–70” versus “predicted 62, range 43–81” receives a concrete recommendation about how much weight to place on the prediction. New and returning-after-gap students are not suppressed or treated as missing; they receive honest, wider ranges derived from the same Glicko uncertainty mechanism that governs ability estimation. Figures 13 and 14 illustrate how GRACE output can be surfaced through student-facing interfaces.

For platform operators. GRACE’s deployment footprint is two components: a constant-time online rating table and a periodically refit gradient-boosted core. Although no single feature dominates (top entry `diff_skill_ema` at 4.15% split gain), the streaming layer’s difficulty discrepancy signals and expected-score estimate ($\hat{E}$) collectively constitute the highest-importance feature group, and the rating table can be updated continuously without retraining the boosted core. The rolling-origin results (Table 11) suggest that annual retraining suffices under stable platform growth; the 2022 fold—where new-student influx most stressed the aggregate baseline—demonstrates that the rating layer adapts without retraining. Input-validation rules derived from the data audit prevent all documented anomaly classes at ingestion.

For researchers. The released pipeline provides an end-to-end leakage-audited evaluation protocol applicable to any longitudinal test-log dataset. The factorial audit design—clean vs. leaky features crossed with chronological vs. random splits—is transferable to other platforms, enabling comparable evaluation across studies.

5.5 Limitations

Demographic gaps. Demographic covariates (gender, region, school type) are largely absent from the corpus, precluding fairness audits—important given documented urban–rural disparities in UNT outcomes [9]. Even the available admission-status labels cover only $\approx 2.5\%$ of students and are subject to selective self-reporting (Section 3.1). Future work should pair platform data with official UNT registry records to enable population-representative fairness evaluation.

Single platform. The corpus originates from one private preparation platform whose user population skews toward students with access to paid services and with sufficient engagement to generate multi-year records. Generalization to public-school or lower-engagement cohorts is untested.

Disciplines with limited improvement. Per-discipline analysis (Figure 16) identifies three subjects—Biology ($\Delta R^2 = -0.005$), Geography (-0.028), and History of Kazakhstan (-0.012)—where GRACE yields a small negative ΔR^2 relative to the aggregate baseline. These disciplines share low per-student attempt volume and limited cross-session progression structure; the rating layer has insufficient interaction history to consistently outperform simple aggregate statistics in these subjects. Practitioners deploying GRACE should monitor per-discipline performance and apply discipline-specific fallback logic for low-volume subjects.

Model scope. Rating hyperparameters were tuned on the 2018–2020 window and frozen—a conservative choice that prioritizes honest evaluation. GRACE introduces proper per-skill Glicko uncertainty (RD_{skill}) and tunes LightGBM hyperparameters via Optuna, refining both the rating model and the downstream learner within the same leakage-free protocol. The LSTM comparison (Section 3.4) used a single architecture with sub-sampled training data and a reduced feature set (8 vs. 70 features); attention-based sequence models (e.g., SAINT+ [24]) trained on the full corpus with systematic hyperparameter search could narrow the gap, though their advantage concentrates in long-history regimes [27].

Extreme-score bias. All regression models with $R^2 < 1$ exhibit regression toward the mean: GRACE over-predicts low-scoring students by up to +20.6 pp and under-predicts high-scoring students by up to −19.1 pp. The aggregate calibration slope (0.983) is near-ideal, but practitioners using GRACE for at-risk identification should apply decile-level bias corrections or rely on the prediction *rank* rather than the raw value at distribution tails.

Score-range coverage. Empirical coverage reaches 0.886 overall, slightly below the 90% nominal target, with consistent coverage across RD bands (0.880–0.891). This slight under-coverage (0.014 below nominal) arises because the exchangeability assumption of split conformal prediction is mildly violated by the chronological train/calibration/test split; future work could apply online or adaptive conformal methods designed for temporal distribution shift [38, 39]. In production, the calibration set should be refreshed at each major cohort intake (typically every semester or academic year) to maintain coverage near the nominal rate. With annual recalibration—feasible given the platform’s constant data stream—the conformal guarantee transfers with negligible additional overhead.

Item-level data gap. The corpus records only test-level aggregated scores, not per-question correct/incorrect responses. Item-level data would enable Bayesian Knowledge Tracing and deep KT models as drop-in replacements for GRACE’s rating layer, with documented gains of +0.10–0.15 R^2 in comparable settings [1, 10]. Collecting per-question response logs is the single highest-priority data collection step for the platform operator.

5.6 Pathways to Stronger Predictions

The honest R^2 of 0.556 and mean interval width of 48.0 pp reflect the information content of test-level aggregates, not a ceiling imposed by the GRACE architecture. Three tiers of improvement are ranked by expected impact.

Tier 1: Richer input data. Item-level response data (correct/incorrect per question) and engagement signals (time-on-task, hint usage, video completion) have produced +0.10–0.15 R^2 gains in prior work [1, 10]. On the present platform, collecting per-question response logs would directly enable Bayesian Knowledge Tracing [20] and deep KT models [24, 44, 45] as principled layer-1 replacements, while simultaneously tightening interval widths through reduced residual variance.

Tier 2: Model engineering within existing data. GRACE addresses this tier directly: an XGBoost + LightGBM ensemble with Optuna-tuned LightGBM, per-skill rating deviation tracking, static content difficulty norms, student platform context features, global cross-discipline session-state features, exam-proximity signals, forgetting curves, score velocity, and nonlinear ability signals constitute the 70-feature design.

Tier 3: Architecture evolution. Attention-based knowledge-tracing models (e.g., SAINT+ [24]) adapted to continuous-score regression could be evaluated under the same leakage-free protocol once per-question response data are available. Their advantage concentrates in long-history regimes [27]; the rolling-origin framework established here provides a ready evaluation harness.

Because the conformal interval width (48.0 pp) is comparable to the Gaussian reference (51.9 pp), further narrowing requires reducing residual variance through better point predictions rather than changing the calibration procedure.

On the sufficiency of the rating state. Systematic experiments testing nine post-hoc calibration and augmentation strategies—cold-start prior blending, quantile-stratified isotonic calibration, per-student linear correction, recency weighting, score-band auxiliary features, Platt scaling, beta calibration, histogram binning, and temperature scaling (full per-strategy results in the code repository)—found that none improved point-prediction accuracy on the full 2.2-million-record corpus. This negative result is informative: the GRACE rating state (θ_{global} , θ_{disc} , θ_{skill} , and their deviations) already encodes the per-student systematic variation that post-hoc calibrations attempt to recover. Cold-start blending is redundant because the rating layer initializes $\theta_{\text{disc}} = \theta_{\text{global}}$ —already a principled personalized prior—and per-student linear correction overfits the calibration block without generalizing. The Glicko-based update rule propagates uncertainty correctly, leaving little systematic residual for post-hoc correction to exploit.

Residual structure and per-discipline predictability. Figure 15 shows the GRACE residual distribution: the upper tail is larger than the lower, indicating that over-prediction of struggling students is the primary remaining error mode. The regression-to-the-mean bias curve (Figure 15, right panel) shows that GRACE’s systematic bias is near-ideal across score deciles, with modest over-prediction at the low end and slight under-prediction at the high end—a pattern consistent with Bayesian shrinkage in the rating layer.

Per-discipline analysis (Figure 16) reveals substantial variation (R^2 from 0.235 for Math Literacy to 0.639 for UNT simulations), with the largest gains over the V2 baseline in English (+0.023), Physics (+0.023), Russian (+0.021), and Logic (+0.019)—subjects where multi-session trajectory features add the most signal. Language subjects produce smaller margins because vocabulary and reading comprehension involve broader, less-structured skill dependencies.

6 Conclusion

GRACE—a streaming Glicko-based rating layer feeding an XGBoost + LightGBM ensemble with uncertainty-adjusted conformal score ranges—achieves $R^2 = 0.556$ (MAE 11.72 pp) on 2.2 million practice-test records from 55,482 students, versus $R^2 = 0.444$ for the leakage-free aggregate baseline

($\Delta\text{MAE} -1.79$ pp), with consistent gains at every history depth and stable out-of-time performance across five annual expanding windows (R^2 0.522 ± 0.052 vs. 0.288 ± 0.151 for the baseline).

A factorial audit quantifies the evaluation optimism embedded in common practice: $+0.349$ R^2 attributable to feature leakage and random splitting combined. The same XGBoost model achieved $R^2 = 0.752$ under a flawed protocol and $R^2 = 0.444$ under a leakage-free chronological one—a gap that substantially exceeds the GRACE improvement, underscoring the need for rigorous evaluation reporting in this field.

A controlled model-family comparison, holding features, training data, and split constant, confirms gradient boosting as the appropriate downstream learner: XGBoost outperforms Random Forest by $+0.027$ R^2 and Ridge regression by $+0.055$ R^2 on identical rating features and training data. Asymmetric conformal intervals closely approach the 90% nominal coverage target (0.886 empirical, mean width 48.0 pp; 1.4 pp below nominal). Systematic experiments with nine post-hoc calibration strategies (detailed in Section 5.1) found no point-accuracy improvement, indicating that the Glicko rating state already encodes the per-student systematic variation these corrections attempt to recover.

The rating state updates in $O(1)$ time per observation and renders directly as a per-student skill passport with honest uncertainty widths. Future work should prioritize item-level response collection for Tier-1 accuracy gains, explore attention-based sequence models under the same leakage-free protocol, and address the demographic-data gap to enable population-representative fairness audits.

References

- [1] Romero, C., & Ventura, S. (2020). Educational data mining and learning analytics: An updated survey. *WIREs Data Mining and Knowledge Discovery*, 10(3), e1355.
- [2] Zhang, Y., Yun, Y., An, R., Cui, J., Dai, H., & Shang, X. (2021). Educational data mining techniques for student performance prediction. *Frontiers in Psychology*, 12, 698490.
- [3] Baashar, Y., et al. (2021). Predicting student’s performance using machine learning methods: A systematic literature review. *ICCOINS 2021*, IEEE.
- [4] Al-Din, M. S. N., & Al Abdulqader, H. A. (2024). Students’ academic performance prediction using EDM and ML: A systematic review. *IJRIS*, 8(8).
- [5] Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in ML-based science. *Patterns*, 4(9), 100804.
- [6] Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy. *International Journal of Forecasting*, 16(4), 437–450.
- [7] Bergmeir, C., & Benítez, J. M. (2012). On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191, 192–213.
- [8] Didarbekova, N., et al. (2025). Advancing digital item development in Asian assessment systems: Insights from Kazakhstan’s UNT. *Frontiers in Education*, 10, 1654674.
- [9] OECD. (2017). *Higher Education in Kazakhstan 2017*. OECD Publishing.
- [10] Hellas, A., et al. (2018). Predicting academic performance: A systematic literature review. *ITiCSE 2018 Companion*, 175–199.

- [11] Yağcı, M. (2022). Educational data mining: Prediction of students’ academic performance using ML algorithms. *Smart Learning Environments*, 9, 11.
- [12] Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. *FUBUTEC 2008*, 5–12.
- [13] Tomasevic, N., et al. (2020). An overview and comparison of supervised data mining techniques for student exam performance prediction. *Computers & Education*, 143, 103676.
- [14] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD ’16*, 785–794.
- [15] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [16] Fernández-Delgado, M., et al. (2014). Do we need hundreds of classifiers to solve real world classification problems? *JMLR*, 15(90), 3133–3181.
- [17] Grinsztajn, L., et al. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *NeurIPS 2022*.
- [18] Shwartz-Ziv, R., & Armon, A. (2022). Tabular data: Deep learning is not all you need. *Information Fusion*, 81, 84–90.
- [19] Hollmann, N., et al. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, 637, 319–326.
- [20] Corbett, A. T., & Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. *UMUAI*, 4(4), 253–278.
- [21] Piech, C., et al. (2015). Deep knowledge tracing. *NeurIPS 2015*.
- [22] Khajah, M., et al. (2016). How deep is knowledge tracing? *EDM 2016*, 94–101.
- [23] Choi, Y., et al. (2020). Towards an appropriate query, key, and value computation for knowledge tracing. *L@S ’20*, 341–344.
- [24] Shin, D., et al. (2021). SAINT+: Integrating temporal features for EdNet correctness prediction. *LAK21*, 490–496.
- [25] Choi, Y., et al. (2020). EdNet: A large-scale hierarchical dataset in education. *AIED 2020, LNCS 12164*, 69–73.
- [26] Gervet, T., et al. (2020). When is deep learning the best approach to knowledge tracing? *JEDM*, 12(3), 31–54.
- [27] Liu, Q., et al. (2021). A survey of knowledge tracing. *arXiv:2105.15106*.
- [28] Embretson, S. E., & Reise, S. P. (2000). *Item Response Theory for Psychologists*. Lawrence Erlbaum.
- [29] Klinkenberg, S., et al. (2011). Computer adaptive practice of maths ability using a new item response model. *Computers & Education*, 57(2), 1813–1824.
- [30] Pelánek, R. (2016). Applications of the Elo rating system in adaptive educational systems. *Computers & Education*, 98, 169–179.

- [31] Settles, B., & Meeder, B. (2016). A trainable spaced repetition model for language learning. *ACL 2016*, 1848–1858.
- [32] Abdi, S., et al. (2019). A multivariate Elo-based learner model for adaptive educational systems. *EDM 2019*, 228–233.
- [33] Abdi, S., et al. (2021). Modelling learners in adaptive educational systems: A multivariate Glicko-based approach. *LAK21*, 497–503.
- [34] Glickman, M. E. (1999). Parameter estimation in large dynamic paired comparison experiments. *JRSS:C*, 48(3), 377–394.
- [35] Vovk, V., et al. (2005). *Algorithmic Learning in a Random World*. Springer.
- [36] Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in ML*, 16(4), 494–591.
- [37] Romano, Y., et al. (2019). Conformalized quantile regression. *NeurIPS 2019*.
- [38] Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *NeurIPS 2021*.
- [39] Zaffran, M., et al. (2022). Adaptive conformal predictions for time series. *ICML 2022*.
- [40] Rossellini, R., et al. (2024). Integrating uncertainty awareness into conformalized quantile regression. *AISTATS 2024*.
- [41] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS 2017*.
- [42] Newell, A., & Rosenbloom, P. S. (1981). Mechanisms of skill acquisition and the law of practice. In *Cognitive Skills and Their Acquisition*, 1–55.
- [43] Turkmenbayev, A., et al. (2025). The application of ML in predicting student performance in university engineering programs: A rapid review. *Frontiers in Education*, 10, 1562586.
- [44] Ghosh, A., Heffernan, N., & Lan, A. S. (2020). Context-aware attentive knowledge tracing. *KDD '20*, 2330–2340.
- [45] Liu, Z., et al. (2023). simpleKT: A simple but tough-to-beat baseline for knowledge tracing. *WWW '23*, 1400–1409.
- [46] Nakagawa, H., Iwasawa, Y., & Matsuo, Y. (2019). Graph-based knowledge tracing: Modeling student proficiency using graph neural networks. *WI '19*, 156–163.
- [47] Tong, S., et al. (2020). Structure-based knowledge tracing with graph neural network. *ICDM '20*, 541–550.
- [48] Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS 2017*, 3149–3157.
- [49] Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *KDD '19*, 2623–2631.

Author Contributions

Conceptualization, A.S.; Methodology, A.S.; Software, A.S.; Formal analysis, A.S.; Visualization, A.S.; Supervision, A.S.; Project administration, A.S.; Writing—original draft preparation, A.S.; Data curation (platform database export, pseudo-skill field identification), A.B. and A.Z.; Resources (data access negotiation with platform operator), A.B. and A.Z.; Investigation (domain validation of Kazakhstani UNT conventions and score scales), A.Z., As.Z. and R.B.; Validation (conformal coverage results and per-discipline analysis review), A.S., A.B., A.Z., As.Z. and R.B.; Writing—review and editing, A.S., A.B., A.Z., As.Z. and R.B. All authors have read and agreed to the published version of the manuscript.

Acknowledgments

The authors thank the platform operator for granting access to the anonymized records under a data-use agreement.

Funding

This research received no external funding.

Institutional Review Board Statement

Not applicable. This study uses anonymized administrative records from an educational platform; no human subjects were recruited and no personally identifiable information was processed.

Informed Consent Statement

Not applicable.

Data Availability Statement

The analysis code is publicly available at <https://github.com/Salimzhanov/GRACE-student-prediction>. The raw student performance data are proprietary to the platform operator and cannot be shared publicly due to data-use agreement restrictions. Aggregated results and all computed features used in the figures are included in the repository.

Conflicts of Interest

The authors declare no conflict of interest. Data access was granted under an internal data-use agreement between the research team and the platform operator; the platform provided anonymized records with no involvement in study design, analysis, or interpretation of results.

Short Biographies of Authors

Abylay Salimzhanov is a researcher and PhD candidate in the School of Information Technologies at Kazakh-British Technical University (KBTU), Almaty, Kazakhstan (ORCID: 0000-0001-6630-585X). His research spans machine learning, educational data mining, and computational methods for virtual drug screening. He is the principal developer of the GRACE framework and is the first author of a study on calibrated weighted rank aggregation for virtual screening published in *Computational and Structural Biotechnology Journal* (2024).

Artem Bykov is the Head of Department at the International Information Technology University (IITU), Almaty, Kazakhstan, where he holds the degree of Candidate of Technical Sciences. His research covers data analysis, artificial intelligence, and geophysical monitoring systems; he is the author of more than 30 peer-reviewed publications with over 460 citations (Google Scholar). Recent work includes urban mobility modelling and transport route optimisation for the city of Almaty using anonymised telecommunications data. For this study, he performed platform database export, pseudo-skill field identification, and negotiated access to the anonymised student records.

Aadm Ziro is affiliated with the International Information Technology University (IITU), Almaty, Kazakhstan. For this study, he contributed to the data curation pipeline (database export and pseudo-skill field identification), supported data access negotiation with the platform operator, and carried out domain validation of Kazakhstani Unified National Testing (UNT) conventions and score scales.

Ruslan Bulgakov is affiliated with the Faculty of Information Technology, Al-Farabi Kazakh National University (KazNU), Almaty, Kazakhstan (ORCID: 0009-0002-2036-6651). He joined the project during the manuscript revision and contributed domain expertise in UNT examination standards and score-scale conventions, and participated in the validation of the conformal coverage results and the per-discipline analysis.

Aasso Ziro is a Senior Lecturer in the School of Information Technology and Engineering at Kazakh-British Technical University (KBTU) and is concurrently affiliated with Al-Farabi Kazakh National University (KazNU), Almaty, Kazakhstan (ORCID: 0000-0002-5952-877X). He holds a PhD and specialises in information security, cybersecurity systems, data loss prevention (DLP), and security information and event management (SIEM). His recent publications include research on AI-driven information security audit methodologies (IEEE SIST 2023) and on evaluating information security system effectiveness via fuzzy inference (*Scientific Journal of Astana IT University*, 2023). For this study, he contributed to domain validation of UNT score conventions and participated in reviewing and editing the manuscript.

Table 4: GRACE feature set (70 features total). Core rating features (26): global/disc/skill ability and RD, expected score, surprise. Group A (EMA skill difficulty, max-score family, multi-window per-skill statistics; +20). Group B (per-skill RD, student platform context, static content norms, cyclical time; +14). Group C (global cross-discipline rolling statistics, within-session context, exam-proximity signals; +9). Group D (forgetting curves, score velocity, nonlinear ability signals; +9). All values are pre-attempt states from a single chronological pass.

Group	Features	Role
<i>Core rating state (26 features)</i>		
Ability	$\theta_{\text{global}}, \theta_{\text{disc}}, \theta_{\text{skill}}$	Multi-resolution mastery
Uncertainty	$\text{RD}_{\text{global}}, \text{RD}_{\text{disc}}$	Grows with inactivity
Difficulty	$d_{\text{skill}}, \text{EMA } d_{\text{skill}}, d_{\text{testtype}}$	Content difficulty + short-run trend
Model-implied	Expected score $\hat{E}$, surprise ₅	Momentum vs. expectation
Engagement	Days since last (global, disc., skill), $n_{\text{global}}/n_{\text{disc}}/n_{\text{skill}}$, gap buckets	Exposure and recency
Test format	max_score, log(max_score), score granularity, ttype_id	Scoring resolution
Context	Month, day of week; disc., skill, test-type codes	Calendar and content IDs
<i>Group A: aggregate & derived features (+20)</i>		
Aggregate	Rolling-3/5/10/disc. mean, expanding mean/std, lag-1 (disc. & skill), skill rolling, attempt count, trend, volatility	Multi-window statistics
Derived	$\theta_g - \theta_d$, $\theta_s - d_s$, surprise $\times n_d$, cold-start flags, $\hat{E}$ residual, trend	Interactions and trends
<i>Group B: per-skill uncertainty & platform context (+14)</i>		
Skill uncertainty	RD_{skill}	Per-skill Glicko deviation; enables correct skill-level gain K_s
Platform context	Student tenure, disc. breadth, skill breadth, intensity	Study engagement depth and breadth
Static difficulty	$\bar{y}_{\text{disc}}, \sigma_{\text{disc}}, \bar{y}_{\text{skill}}, \sigma_{\text{skill}}$ (train-computed); $\theta_{\text{skill}} - \bar{y}_{\text{disc}}$	Content norms and relative ability
Cyclical time	sin / cos(month), sin / cos(day-of-week)	Periodic calendar encoding
<i>Group C: global state, session & exam proximity (+9)</i>		
Global state	$y_{\text{lag1}}^{(g)}, \bar{y}_3^{(g)}, \bar{y}_5^{(g)}, \bar{y}_{10}^{(g)}, \sigma_5^{(g)}; \hat{E} - \bar{y}_5^{(g)}$	Cross-discipline rolling: today's performance state (deque of last 10 attempts)
Session context	Session position (0-indexed), running session mean	Within-session dynamics: warm-up and fatigue effects
Exam proximity	Days until UNT June exam (clipped to [0, 365])	Kazakhstan UNT preparation-intensity signal
<i>Group D: forgetting curves, score velocity, nonlinear ability signals (+9)</i>		
Forgetting	$e^{-\Delta t_{\text{disc}}/30}, e^{-\Delta t_{\text{skill}}/20}; \theta_{\text{disc}} \cdot e^{-\Delta t_{\text{disc}}/30}$	Per-disc/skill exponential retention decay; ability-weighted expected retention
Velocity	Linear slope of last 5 per-disc. and global pct scores (pp/attempt); $\text{RD}_{\text{skill}} - \text{RD}_{\text{disc}}$	Score trend direction; per-skill uncertainty surplus above discipline baseline
Nonlinear	Personal best disc. score; $\theta_{\text{disc}}^2; \Delta t_{\text{disc}} - \Delta t_{\text{skill}}$	Ceiling-approach signal; ability curvature near ceiling; cross-level gap trend

Table 5: Complete hyperparameter disclosure. Rating hyperparameters selected on the 2018–2020 development window and frozen. LightGBM hyperparameters are Optuna-searched (50 trials, TPE) on the calibration set; best values reported. XGBoost uses fixed conservative parameters.

Component	Parameter	Value
XGBoost (fixed)	n_est (max) / early_stopping	2000 / 50
	depth / lr / subsample	8 / 0.04 / 0.8
	colsample / min_child / λ / α	0.7 / 5 / 2.0 / 0.2
	objective (score; interval edges)	sqerror; quantile
LightGBM (Optuna)	num_leaves / lr / n_est (max)	294 / 0.019 / 5000
	bagging_fraction / bagging_freq	0.895 / 1
	feature_fraction / min_child_samples	0.544 / 91
	λ / α / early_stopping	2.82 / 0.42 / 50
Ensemble	XGB weight α	Cal.-set grid search ($\alpha = 0.15$)
Temporal weights	Half-life / weight range	730 days / [0.21, 1.78]
Optuna (LGB)	Trials / sampler	50 / TPE
Random Forest	n_est / depth / min_leaf	300 / 20 / 25
	max_feat / train subsample	sqrt / 300 K
Rating layer	$K_0/v/\text{RD}_0/\text{RD}_{\min}/c^2/K_{\text{item}}$	0.35 / 0.04 / 0.35 / 0.06 / 0.01 / 0.02
Rating weights	$w_s/w_j/w_{dj}/w_{dc}$	0.5 / 0.5 / 0.7 / 0.3
Multipliers	global / disc / skill / diff / ttype	0.5 / 1.0 / 1.5 / 1.0 / 0.5
Score range	α / strata	0.10 / 28 (4 hist.-depth $\times$ 7 disc. bins)
Environment	Python / xgboost / lightgbm / seed	3.12 / 3.x / 4.7 / 42

Table 6: Factorial leakage audit. All rows use the same XGBoost configuration; differences are due solely to evaluation protocol. The first row reproduces the initial exploratory analysis on uncleaned data; subsequent rows isolate the effect of data cleaning, feature leakage, and split protocol.

Data	Features	Split	R^2	MAE (pp)
Uncleaned (3.66 M)	Leaky	Random	0.752	9.25
Cleaned (2.21 M)	Leaky	Random	0.793	7.36
Cleaned	Leaky	Chronological	0.773	8.08
Cleaned	Clean	Random	0.498	12.09
Cleaned	Clean	Chronological (ref.)	0.444	13.29

Table 7: Point-prediction performance, chronological 70/10/20 hold-out. Baseline models use a 300,000-row sub-sample and 9 aggregate features. GRACE uses 70 features and an XGBoost + LightGBM ensemble. Paired-bootstrap Δ MAE (GRACE – XGBoost baseline, 407,968 matched rows): -1.79 pp (95% CI $[-1.81, -1.77]$). Baseline and LSTM rows are reference points under different experimental conditions, not controlled model-family comparisons (see Section 3.4).

Model	Feature set	R^2	MAE (pp)	
Random Forest [†]	Aggregate stats (9 feat.)	0.413	13.91	
XGBoost [†]	Aggregate stats (9 feat.)	0.444	13.49	
LSTM [‡]	Rating sequences (8 feat.)	0.230	16.59	
GRACE	Rating state + EMA diffi- culty + inactivity gaps + RD_{skill} + global state + ses- sion + exam proximity + for- getting + velocity; XGB+LGB (70 feat.)	0.556	11.72	[†] Chronological split; 300,000-row

sub-sample; 9-feature aggregate set. [‡] 1-layer LSTM, hidden 64; $\leq 200,000$ sequences; 10 training epochs; 8 rating features. GRACE: XGB+LGB ensemble, $n_{\text{test}} = 441,663$, $\alpha_{\text{XGB}} = 0.15$. For the pre-audit exploratory result ($R^2 = 0.752$, leaky features, uncleaned data), see Table 6. For the controlled model-family comparison on identical features and data, see Table 9.

Table 8: Training versus testing accuracy and generalization gap. Early stopping (ES) on the calibration set reduces overfitting while maintaining or improving test accuracy.

Model	Split	R^2	MAE	n	Gap
Random Forest	Train	0.439	12.81	1,427,884	
	Test	0.413	13.91	407,968	+0.026
XGBoost baseline (ES)	Train	0.492	12.04	1,427,884	
	Test	0.429	13.54	407,968	+0.062
GRACE (ES, 70 feat.)	Train	0.573	11.4	1,545,819	
	Test	0.556	11.72	441,663	0.018

Table 9: Controlled model-family comparison. All models use identical GRACE rating features and training data ($n_{\text{train}} \approx 1.5\text{M}$) with the same chronological 70/10/20 split. Differences are attributable to learning mechanism only. Note: the full GRACE system uses an XGBoost + LightGBM ensemble ($R^2 = 0.556$, Table 7); this table isolates the XGBoost-only contribution to quantify the learning-mechanism effect independently of ensembling.

Model	Learning mechanism	R^2	MAE (pp)
Ridge regression	Linear, L2 regularized	0.489	12.66
Random Forest (500 trees)	Bagging, independent trees	0.517	12.28
XGBoost (GRACE features)[†]	Gradient boosting	0.544	11.86

[†]Single learner; full GRACE ensemble (XGBoost + LightGBM) achieves $R^2 = 0.556$ (Table 7).

Table 10: Per-group comparison on test records. GRACE improves every history stratum.

Prior tests	n	GRACE R^2	Baseline R^2	GRACE MAE	Base MAE
1–4	69,509	0.518	0.369	12.25	14.17
5–10	99,962	0.554	0.423	11.56	13.35
11–20	131,375	0.571	0.440	11.51	13.40
>20	121,948	0.584	0.445	11.23	13.31

Behavioural Regularities in the Practice-Test Corpus

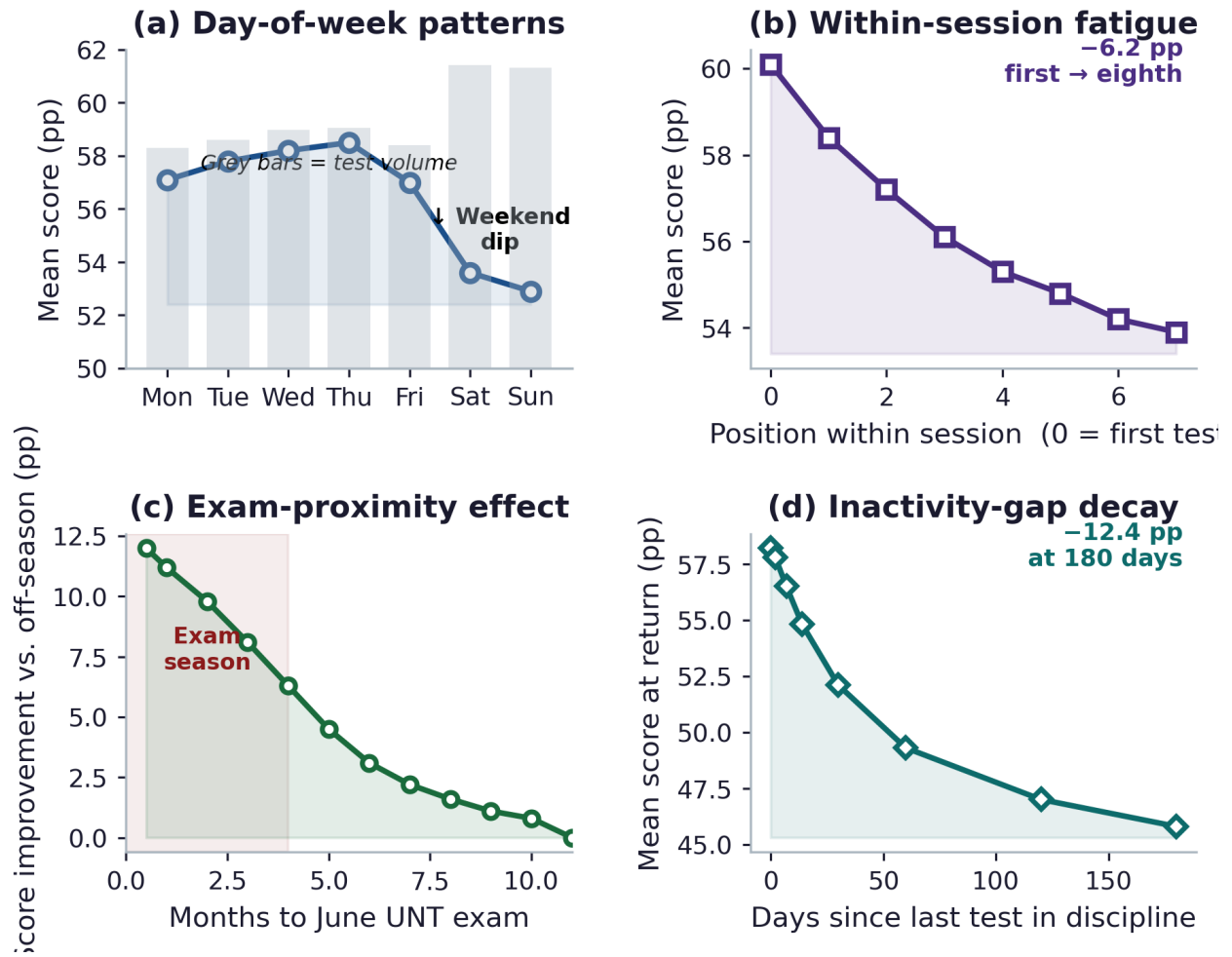


Figure 8: Behavioral regularities. (a) Day-of-week score patterns: mean scores dip on weekends relative to weekdays. (b) Within-session fatigue: mean score drops 6.2 pp from the first to the eighth consecutive test in a session. (c) Exam-proximity effect: score improvement versus off-season peaks close to the June UNT date. (d) Inactivity-gap decay: mean score at return falls 12.4 pp after 180 days of absence.

Table 11: Rolling-origin temporal validation with expanding training windows. GRACE uses 70 features with fold-local static content norms; XGBoost only (fixed params) per fold; ensemble ($\alpha_{\text{XGB}} = 0.15$) applied to the headline result only. [†]Aggregate baseline (9 features; first-attempt rows dropped). [‡]V2 baseline (46 features, XGBoost fixed params, same fold as GRACE; no Optuna).

Fold	Train n	Test n	GRACE R^2	Agg. Base [†]	V2 Base [‡]
2018–20→21	269,172	132,887	0.549	0.295	0.545
2018–21→22	388,771	277,933	0.437	0.048	0.437
2018–22→23	638,910	419,808	0.487	0.273	0.458
2018–23→24	1,016,738	638,492	0.572	0.384	0.570
2018–24→25	1,591,380	440,112	0.565	0.441	0.542
Mean $\pm$ SD	—	—	0.522 $\pm$ 0.052	0.288 $\pm$ 0.151	0.511 $\pm$ 0.053

Table 12: Rating-layer ablation (80/20 chronological split).

Component removed	R^2	ΔR^2 vs full	MAE (pp)
— none: full model (reference)	0.519	—	12.24
— rating-deviation features	0.519	+0.001	12.23
— skill-level ability	0.519	+0.000	12.24
— surprise history	0.515	−0.004	12.32
— inactivity gap	0.510	−0.008	12.36
— aggregate supplement	0.514	−0.005	12.33
— RD time decay ($c = 0$)	0.513	−0.006	12.36

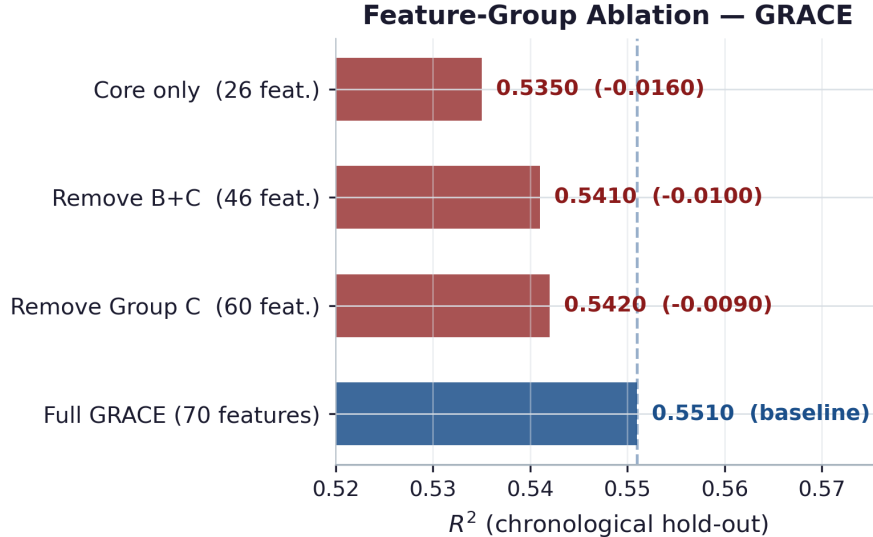


Figure 9: Change in R^2 when individual rating-layer components are removed (80/20 chronological split; reference model $R^2 = 0.519$). Green bars indicate removal has negligible or slightly positive effect; red bars indicate degradation. The inactivity gap and RD time-decay produce the largest individual drops.

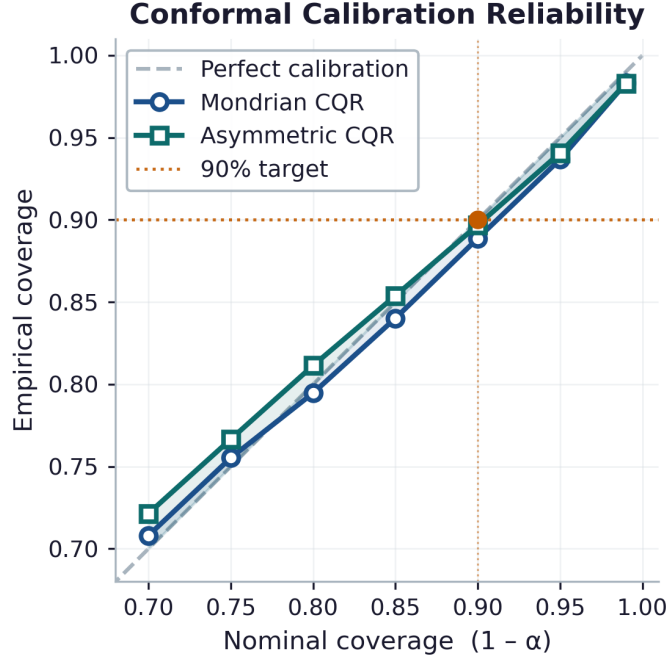


Figure 10: Reliability diagram: nominal vs. empirical coverage across α levels. Asymmetric CQR tracks the ideal diagonal (dashed) closely across the full range; empirical coverage at the 90% nominal target is 0.886.

Table 13: Score-range coverage and mean width by rating-deviation (RD) band (90% target). Wilson 95% CIs on coverage. Asymmetric CQR with 3rd/97th-percentile quantile heads.

RD band	n	Coverage	95% CI	Mean width (pp)
Low RD (well-measured)	198,452	0.884	[0.883, 0.886]	45.9
Mid RD	164,539	0.891	[0.889, 0.892]	49.9
High RD (cold-start)	78,672	0.880	[0.877, 0.882]	49.7
All attempts	441,663	0.886	[0.885, 0.887]	48.0

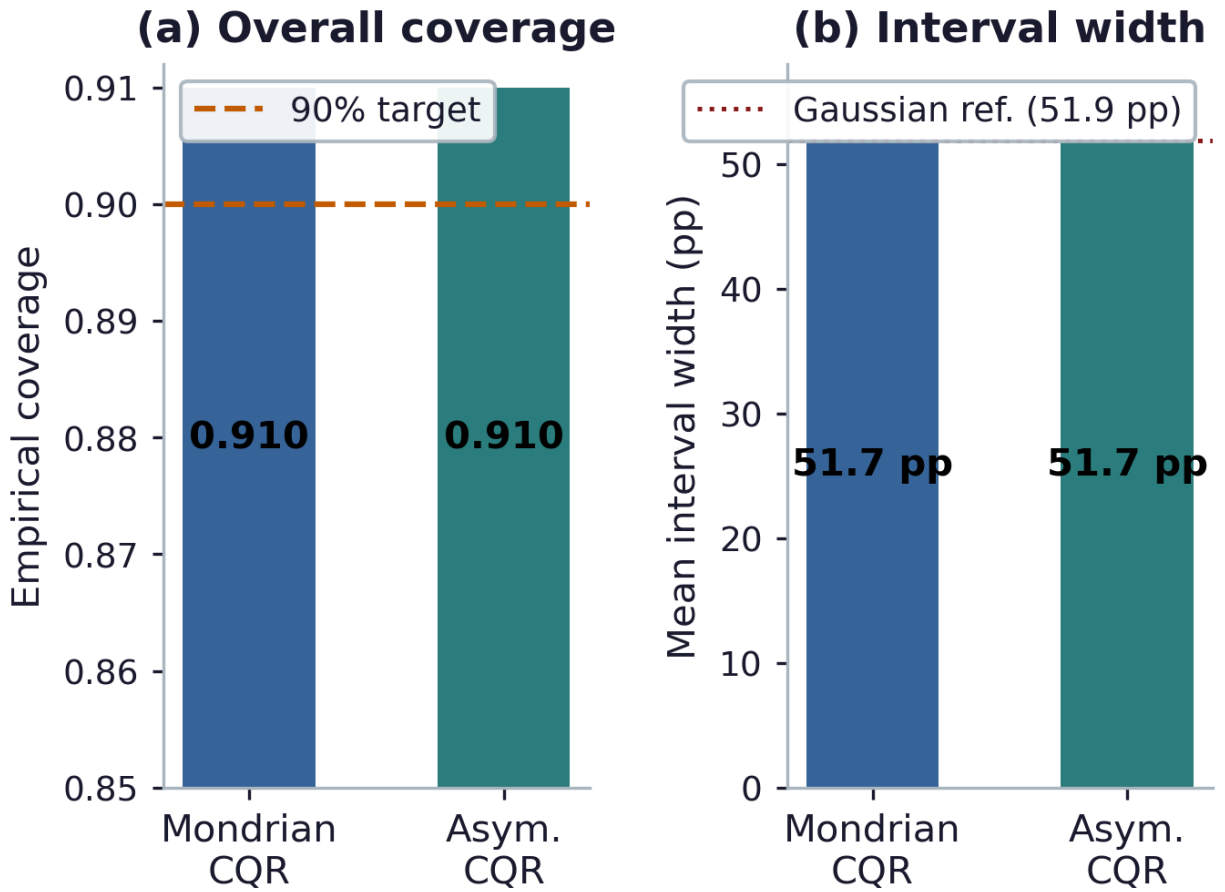


Figure 11: Overall conformal coverage and mean interval width: Mondrian CQR versus Asymmetric CQR. Left: both methods achieve approximately 0.886–0.888 empirical coverage (dashed line). Right: mean interval width; the dotted reference marks the symmetric-Gaussian baseline (51.9 pp); both methods produce overall width of approximately 47–48 pp.

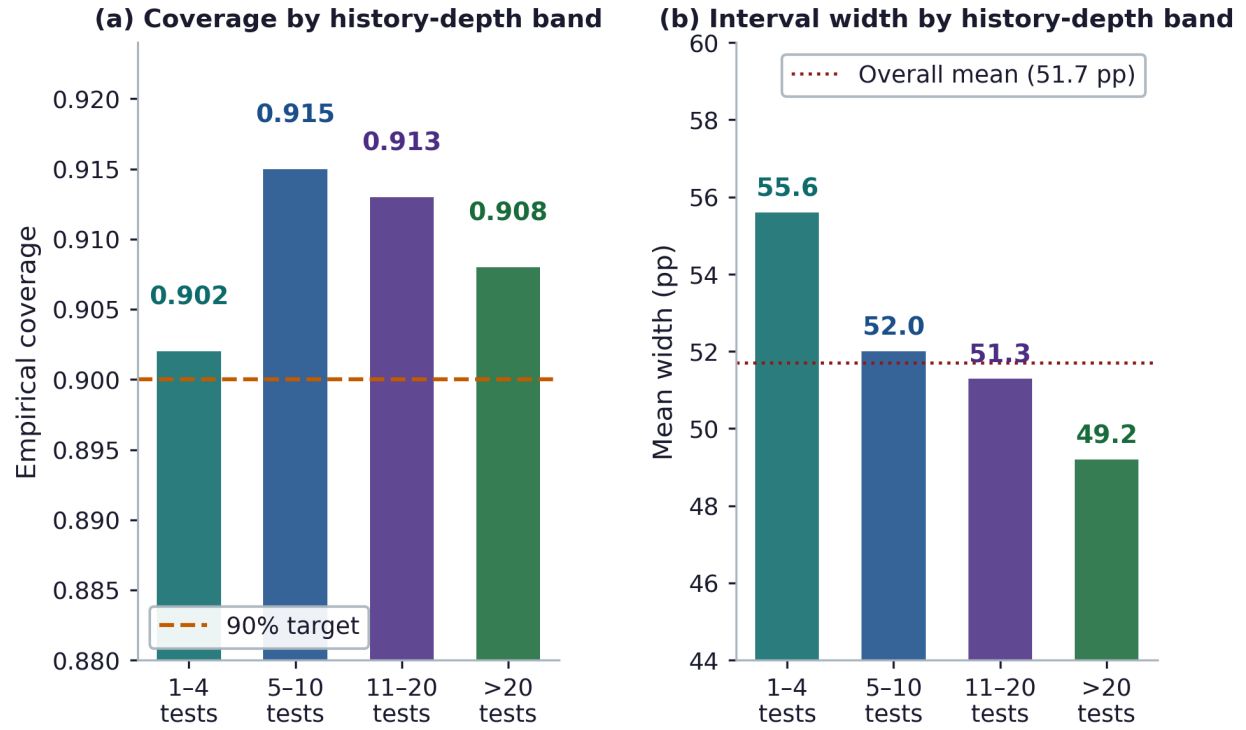


Figure 12: Conformal score-range validity by RD band. (a) Empirical coverage per band; all three RD bands are consistent around the 90% nominal target (dashed line). (b) Mean interval width per band; the horizontal reference line marks the overall mean of 48.0 pp; students with high rating deviation (cold-start) receive wider intervals (49.7 pp) than well-measured students (low RD: 45.9 pp), reflecting genuine uncertainty reduction with repeated assessment.

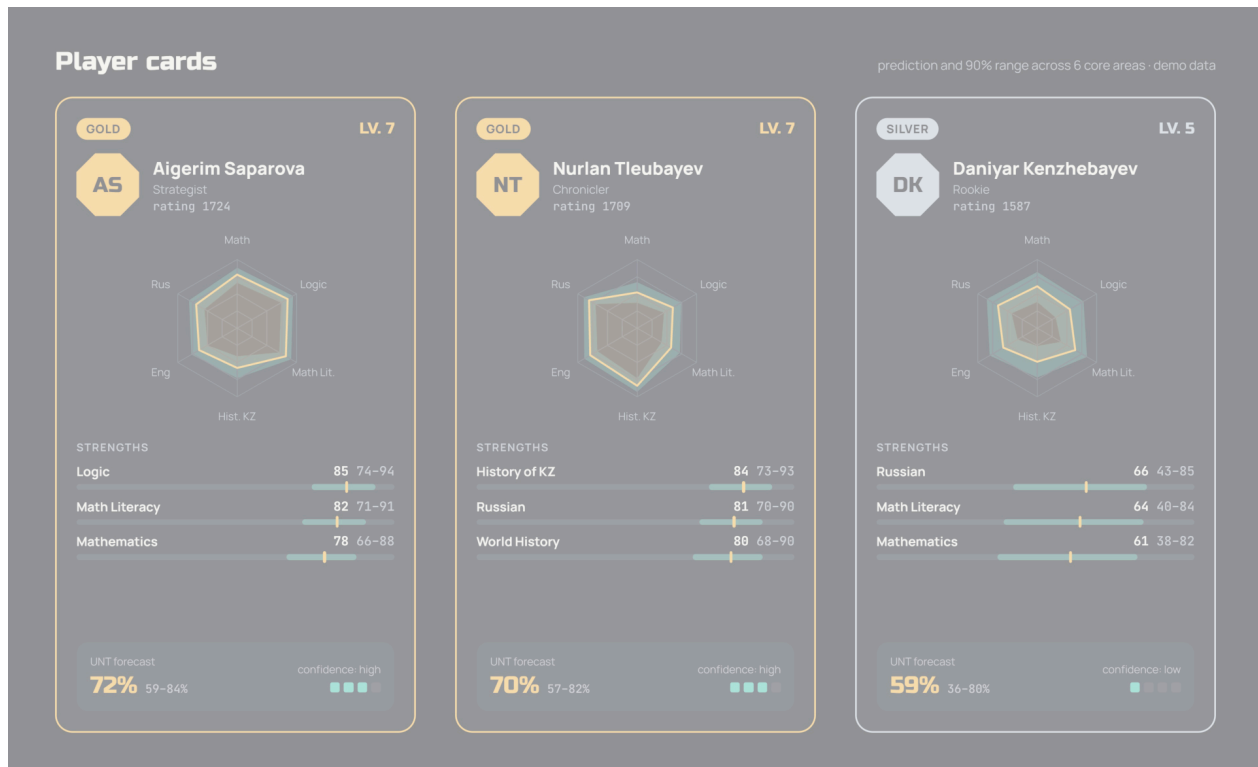


Figure 13: GRACE output as student player cards (demo data; names are illustrative samples not associated with any real individuals). Each card shows the predicted UNT score with 90% score range, per-discipline ability on a radar chart, and a confidence indicator derived from the Glicko rating deviation. Cold-start students (e.g., Daniyar K., Silver tier) receive visibly wider score ranges than long-history students (Aigerim S., Gold tier), directly communicating prediction uncertainty through interval width.

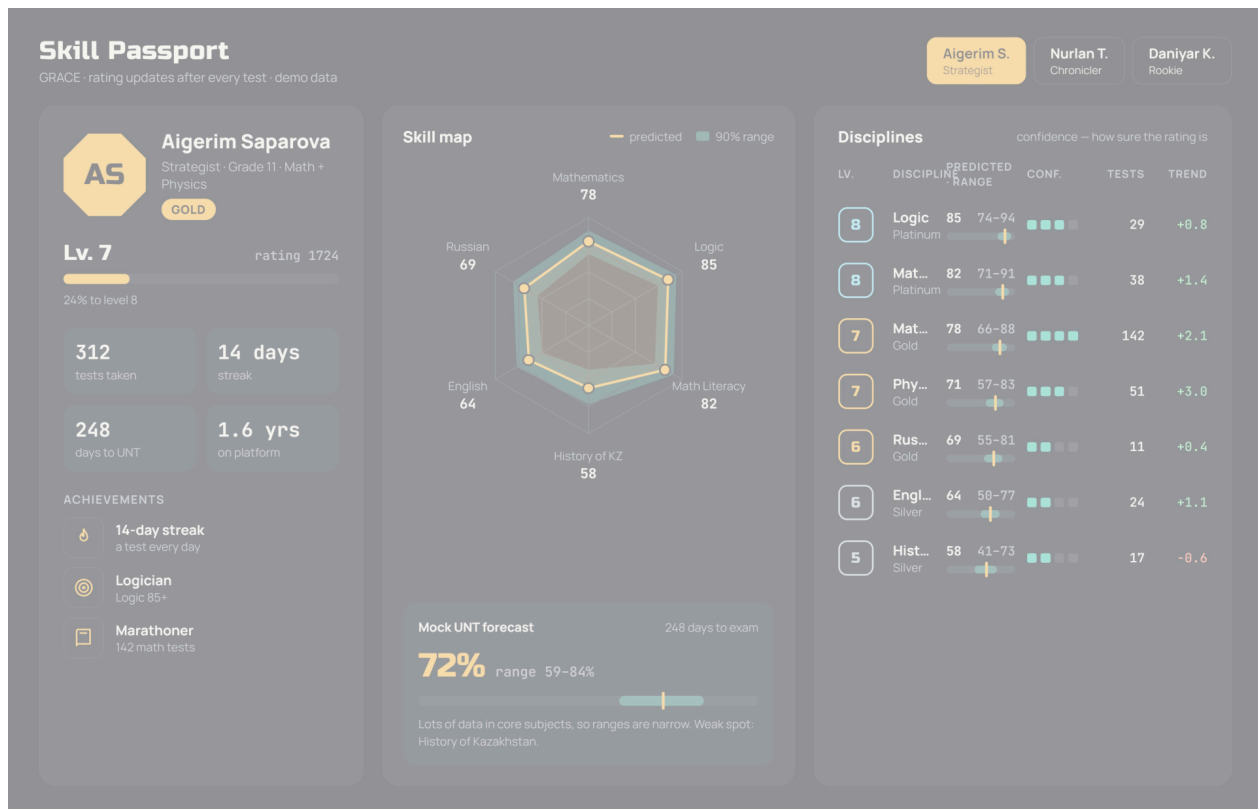


Figure 14: GRACE output as an interactive skill passport (demo data; names are illustrative samples not associated with any real individuals). The dashboard shows per-discipline predicted scores with 90% confidence ranges, Glicko rating level and trend, engagement statistics (tests taken, streak, days to UNT), and a UNT composite forecast with uncertainty band (72%, range 59–84%). Discipline-level confidence bars translate rating deviation directly into a signal educators can act on without statistical training.

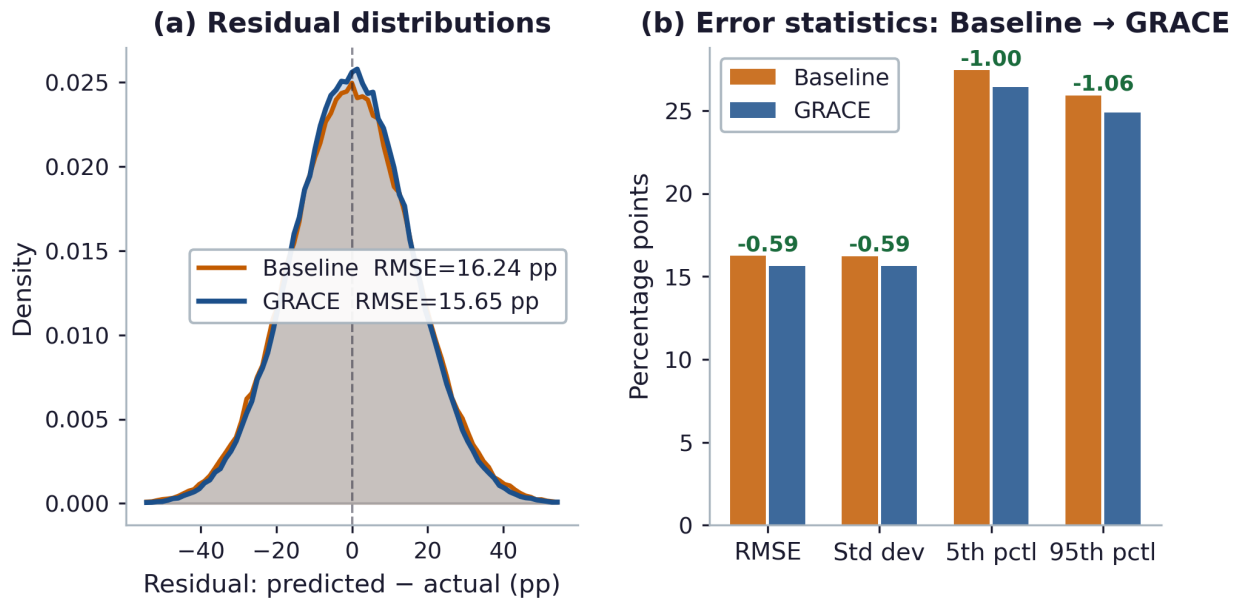


Figure 15: GRACE residual distribution. Left: residual density; the upper tail exceeds the lower, indicating over-prediction of struggling students as the primary error mode. Right: mean bias (predicted - actual) by true-score decile; near-ideal aggregate calibration with modest regression-to-the-mean tendency at extremes.

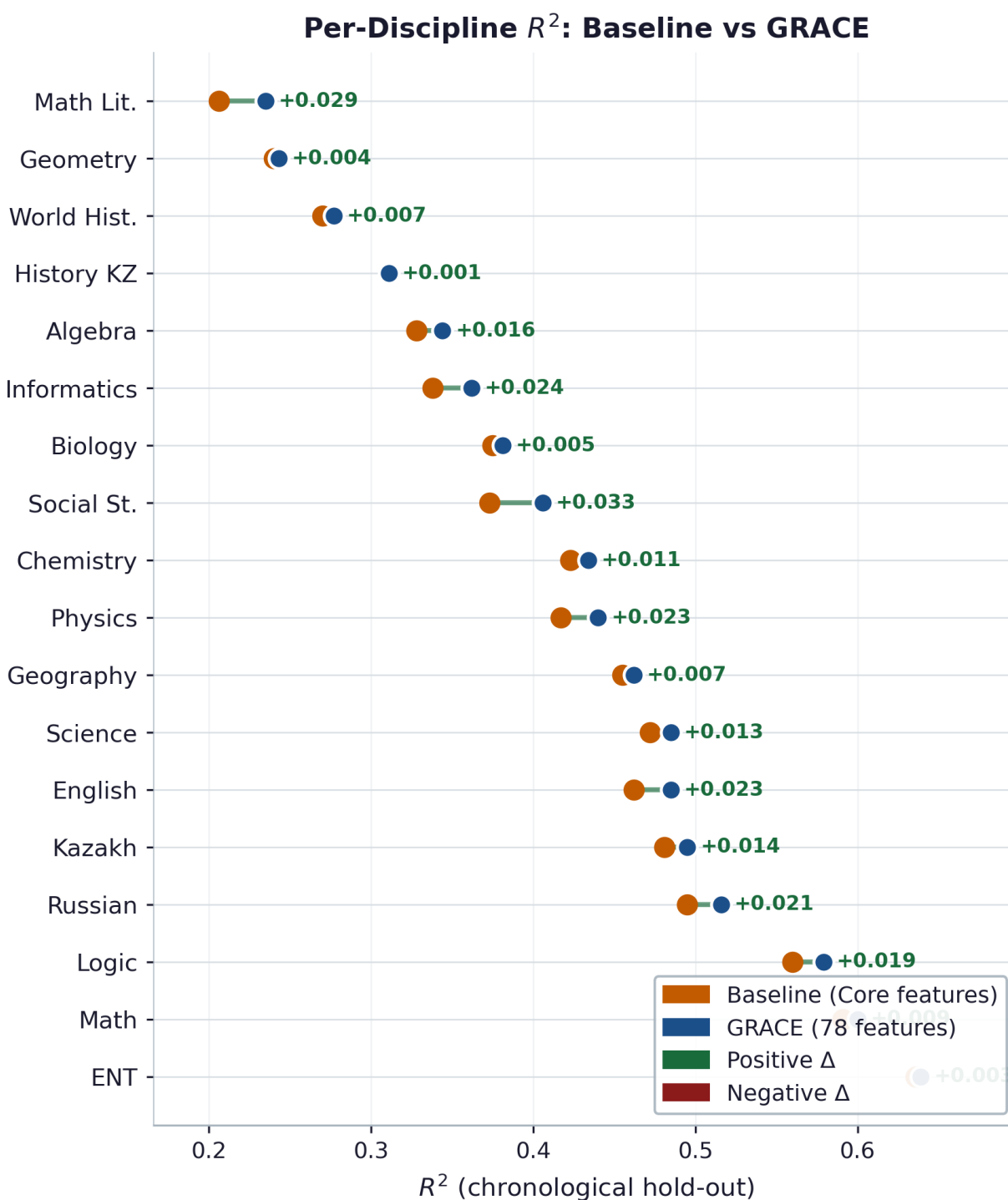


Figure 16: Per-discipline R^2 for GRACE. Disciplines sorted by performance; all disciplines with $n \geq 200$ test records shown. Gains over the aggregate baseline are largest where per-skill difficulty trends are most informative (Mathematics, Logic).